



PDMS with designer functionalities—Properties, modifications strategies, and applications

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ABSTRACT

Rapid progress in micro- and nanotechnologies such as lab-on-a-chip (microfluidic networks, sensors, actuators, and connectors), soft lithography (replica moulding, microcontact printing and affinity contact printing), and stretchable transparent electronics has strongly benefitted from high-performance polymers like poly(dimethyl)siloxane (PDMS) that are suited for high-fidelity microsystem construction and rapid prototyping. While basic PDMS has been a unique enabling material, recent progress in tailoring PDMS to specific requirements will render this material even more valuable in the future.

Basic PDMS is elastic, transparent, biocompatible, gas-permeable, and forms conformal contact with surfaces.

Surface modifications of PDMS, inducing properties such as hydrophilicity, electrical conductivity, anti-fouling, energy harvesting, and energy storage (supercapacitors) are of major interest.

Bulk modifications can alter PDMS properties such as elasticity, electrical and thermal conductivity. Such bulk modified PDMS composite materials can be created by embedding free molecules (e.g., dyes), nanoparticles (graphene, carbon nanotubes, and various other of organic and inorganic nature) or microparticles, or by altering the composition of the prepolymers before polymerization.

Both, surface and bulk modifications of PDMS open avenues to a multitude of tuneable characteristics optimized for a diverse set of applications ranging from integrated micro- (lab-on-a-chip) to macro-systems (biomedical devices and epidermal electronics).

In microfluidic systems design exploiting modified PDMS, a key aspect of this review, the unique features of these materials permit rapid, easy, and reproducible construction without the need for elaborate facilities and trained personnel as compared to other materials like silicon.

This review focuses on recent progress in modification strategies to alter deliberately PDMS surface and bulk properties, especially for microfluidic, biological, flexible electronics, e-skin, and self-healing applications.

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Abbreviations: AgNW, silver nanowires; APTES, 3-aminopropyltriethoxysilane; ATRP, atom transfer radical polymerisation; BSA, bovine serum albumin; CNT, carbon nanotubes; CuNW, copper nanowires; CVD, chemical vapour deposition; FITC, fluorescein isothiocyanate; GO, graphene oxide; LBL, layer-by-layer; MAM, mechanically assembled monolayer; MWCNT, multi-walled carbon nanotubes; NHS, *N*-hydroxysuccinimide; PDA, polydopamine; PEG, polyethylene glycol; SWCNT, single-walled carbon nanotubes; TRITC, tetramethylrhodamine isothiocyanate; WCA, water contact angle; XPS, x-ray photoelectron spectroscopy.

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1. Introduction

In general, polymers with a $-R_2Si-O-$ unit are termed silicones, while the $-Si-O-$ repeat unit is also called siloxane. The strength of the Si–O bond gives the polymer its thermal and chemical stability, which is important for its use in high-temperature applications [1,2]. In PDMS, the flexibility of the siloxane backbone permits the chains to easily arrange and rearrange themselves so as to place the methyl groups at their surfaces and interfaces [3]. Several silicone elastomers are available, with Sylgard 184® from DOWSIL being the most cited in the scientific literature in lab-on-a-chip devices [4,5] within the last two decades [6–12]. The composition of Sylgard 184® is not disclosed by the manufacturer. However it is reported to be as follows: Base: dimethylsiloxane oligomers with vinyl end groups (>60%), silica filler (dimethylvinylated and trimethylated silica, 30–60%), tetra(trimethylsiloxy) silane (1–5%) and ethylbenzene (<1%), and a platinum catalyst. Curing agent: a cross-linking agent (dimethyl methylhydrogen siloxane, 40–70%) and an inhibitor (tetramethyl tetra vinyl cyclotetrasiloxane 1–5%). The composition of another PDMS used in microfluidics, RTV-615 from Momentive Materials, is Base: polyvinylsiloxane (60–90%), polyalkylalkenylsiloxane (10–30%), and a catalyst. Cross-linking agent: polyvinylsiloxane (30–60%), modified silica (SiH) (30–60%), octamethylcyclotetrasiloxane (<1%) and toluene (<1%) [13]. Elastomer fillers such as silica fillers are added to base polymers to increase elastic modulus. A reaction scheme of PDMS polymerized by platinum catalysts is presented in Fig. 1 (cross-linking by

Table 1

Some relevant characteristics of PDMS (Sylgard 184, 1:10 catalyst to base polymer).

Characteristic	Value	References
Optical transparency	240–1100 nm	[16–18]
Surface tension	21–22 mN/m	[1]
Glass transition temperature	–125 °C	[1]
Electrical conductivity	$2.9 \times 10^{-14} \Omega^{-1} \text{cm}^{-1}$	[22]
Heat conductivity	0.27 W/m·K	[22]
Young's elastic modulus <i>E</i>	~1–3 MPa	[19,43]
Surface tension	21 mN/m	[1]
Water vapour diffusion coefficient	~1000–6000 $\mu\text{m}^2 \text{s}^{-1}$	[41]
Oxygen diffusion coefficient	~2000–4000 $\mu\text{m}^2 \text{s}^{-1}$	[49]
CO ₂ diffusion coefficient	~1000 $\mu\text{m}^2 \text{s}^{-1}$	[50]

addition). In larger volume applications in industry and household, other, inexpensive polymerization chemistries for siloxanes, e.g. cross-linking by condensation, is widespread. These materials are also called one-part room temperature vulcanization (RTV) sealants, although requiring moisture as a second component. However, the release of acetic acid during polymerization limits the usefulness of such chemistry for sensitive and biomedical applications [14].

Some physical properties of PDMS are listed in Table 1. PDMS is optically transparent (240 – 1100 nm) [16–18], is biocompatible [13] forms conformal contact (van der Waals contact or molecular contact) and has a low surface free energy (is hydrophobic; contact angle with water ~110°). It has a unique flexibility with a Young's

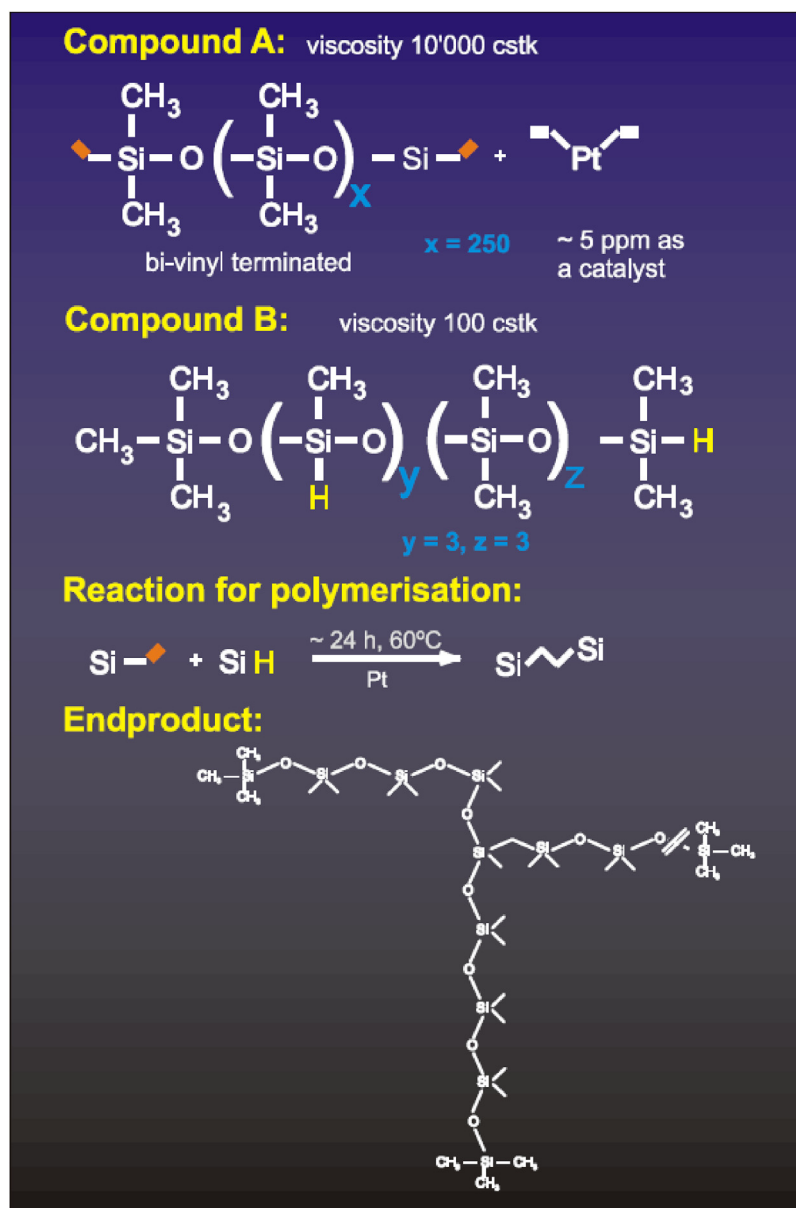


Fig. 1. Reaction educts and polymerization reaction of poly(dimethyl)siloxane (PDMS) [15].

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elastic modulus of $E \approx 1\text{--}3 \text{ MPa}$ [19] due to one of the lowest glass transition temperatures of any polymer ($T_g \approx -125^\circ\text{C}$). It is heat resistant (thermal degradation at temperatures $>400^\circ\text{C}$ [20,21]), thermally insulating ($\sim 0.27 \text{ W m}^{-1} \text{ K}^{-1}$) [22], electrically insulating (breakdown voltage, $2 \text{ E}10^7 \text{ V/m}$), has a high gas permeability, a low dielectric constant (2.7), displays very low surface tension values (around $21\text{--}22 \text{ mN/m}$), and a very low solubility parameter of $15.5 \text{ (J/cm}^3)^{1/2}$. PDMS exhibits a low change in the shear elastic modulus G ($\approx 250 \text{ MPa}$) versus temperature ($1.1 \text{ kPa } ^\circ\text{C}^{-1}$) [3,23], and it can be produced at industrial scale [18,24–27]. Main advantages of PDMS for the end user are its low cost, its rapid prototyping capability by micromoulding, and its propensity to seal to other planar surfaces without the necessity of using adhesives. Also, it is not brittle, as compared to glass or silicon devices, but it might break upon strong bending and alter channel geometry (and therefore also fluidic resistance) when the pressure inside a microdevice becomes too high.

PDMS in its unmodified form also has limitations, depending on its intended use. For example, its hydrophobicity can be a problem when attempting to fill microchannels *via* capillary force [28]. Surface treatment with oxygen plasma can temporarily render it into a hydrophilic surface [18]. Control of electro-osmotic flow is difficult [29]. Protein adsorption (e.g. leading to protein depletion inside a microfluidic channel) may be an issue when trying to quantify low concentrations of analyte [6]. The permeability of PDMS for most organic solvents limits its use due to solvent diffusion into the bulk thereby inducing swelling [30]. PDMS is permeable for water vapour, potentially leading to changes of the concentration of electrolytes in cell culture media and thereby leading to disturbances of cellular growth or cell death [13]. It is permeable for small lipophilic molecules (e.g. Nile red, oestrogen, antibiotics, anti-cancer molecules, retinoic acid, or vitamin D3). This is a problem when they act as intercellular signal in microscale cell culture or microphysiological systems (also termed 'organs-on-chip', 'tissue-

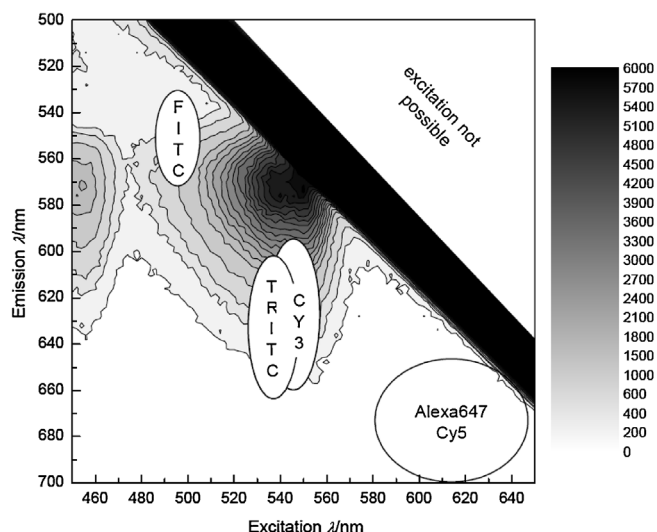


Fig. 2. Fluorescence spectrum of Sylgard 184. While both FITC and TRITC are within the autofluorescence of PDMS, that is not the case for either Alexa Fluor 647 or Cy5. The black bar corresponds to saturation of the detector (>65,000 counts) [39]. Copyright 2004. Reproduced with permission from the Royal Society of Chemistry.

chips'), should be quantified, e.g. in electrophoresis or single cell culture experiments, or should be tested for optimal therapeutic concentration [31–34]. It was shown, that the absorption of molecules by PDMS is related to the logarithm of the octanol–water partition coefficient ($\log P$) of the molecule [31]. Sol-gel methods [35] and parylene coatings [36] were proposed to abrogate lipophilic molecule absorption. PDMS oligomers may diffuse from the bulk into contacting aqueous solutions, such as cell culture medium. In fact, even after Soxhlet extraction in ethanol, a wide range of oligomer lengths was detected in microchannels by mass spectrometry. The oligomers were also present in the plasma membranes of cells cultured on PDMS, as evidenced by detection of silicon atoms via energy-dispersive X-ray spectroscopy [32]. However, the biological implications of this phenomenon have not yet been completely elucidated (see also chapter 3.2.5 Cell culture below) [13,37]. Similarly, uncrosslinked oligomers were shown to migrate to the surface of moving aqueous droplets on a vertical PDMS surface, leading to a second, faster speed regime when the surface of the droplets is completely covered by oligomers [38]. PDMS was reported to be weakly fluorescent in a certain spectral range (e.g. in the FITC and TRITC, but not in the Alexa Fluor® 647 excitation range [excitation e.g. with a 633 nm helium–neon laser]). See Fig. 2 [39].

A comparison of PDMS (the standard in microfluidic cell culture [40]) with polystyrene (the standard in conventional macro-scale cell culture) was done, listing challenges and strategies to overcome certain limitations of PDMS. For example, Parylene coating was suggested for overcoming permeability to a) water (vapour) [41,42] and b) small hydrophobic molecules (as mentioned above) [42], and c) leaching of non-crosslinked oligomers into the contacting aqueous phase [43]. PDMS is biodegradable in the soil through abiotic hydrolysis to monomeric dimethylsilanediols and through other mechanisms such as microbiotic degradation and volatilization into CO_2 and silica [44]. PDMS is also biodegradable *in vivo*, i.e. underneath biofilms and in an acidic environment (e.g. gastric acid with pH ~ 1). PDMS can be used to encapsulate supercapacitors [45], functional devices [46], and bioelectronics [47]. Multi-cyclic load-deflection curves of thin supported PDMS membranes (down to 70 nm thickness with dilution in hexane) were measured, induced by application of unilateral pressure. Their

applicability to mechanical and chemical sensors on a micro/nano-scale is proposed [48].

These achievements delineate ways to overcome or modify limitations of the permeability properties of PDMS. A vast amount of research has been dedicated to overcome all these limitations via specific bulk and surface modifications, some of which is presented in this previous review [43].

Soft lithography [51,52] has become wide-spread since its invention and comprises (i) replica moulding which allows the rapid and facile (e.g. no cleanroom required) production of PDMS with a desired surface structure by simply casting it over a suitable mould, followed by polymerisation – in contrast to glass or silicon structures that require expensive infrastructure, trained personnel for production and break easily [18],

(ii) microcontact printing [53] and affinity contact printing [54], which are based on the flexibility of PDMS and its propensity to form conformal contact with other surfaces, and

(iii) rapid prototyping, i.e. the reduced time needed to form a surface structure from a digital (CAD) master, e.g. via replica moulding [55]. Solving a limitation of soft lithography, a method termed 'PDMS photolithographic surface micromachining' was invented, which allows the generation of through holes in thin PDMS films (down to 100 nm) down to $\sim 2 \mu\text{m}$ diameter. It relies on direct reactive ion etching of PDMS with SF_6 and O_2 , which however requires dedicated infrastructure and is more hazardous [56].

This review focuses on techniques for tailoring of PDMS properties to application-specific requirements by surface and bulk modifications, with special emphasis on published work related to microfluidics since 2012. Several reviews on surface modifications for microfluidics were published in the years from 2003 to 2012 [57–60]. This review is divided into various sections, based on modification strategies and properties to be modified. When properties of a modified PDMS fits to several sections the most salient feature is taken as criterion for assignment to a section. Applications are mentioned where relevant for discussion of specific modifications. Some discussed modification strategies are cited because they could potentially be used in a microfluidic design in the near future even though they have not yet found broad application. Moreover, the field of PDMS-based electronics is not covered exhaustively due to the vast amount of literature appearing, dealing with this burgeoning very active field, which is beyond the scope of this article. A short overview on wearable electronics is presented in chapter 3.5.

2. Tailoring bulk properties

Bulk properties can be tailored by a) variation of curing conditions [19], b) mixing different ratios of base with cross-linker (e.g. 5.7–42.9 wt.% [61]), c) by mixing different types of PDMS [62], d) by inclusion of other molecules, or e) inclusion of fillers. Fillers are small particles and can be classified into the three groups carbon based fillers, metallic fillers, and ceramic fillers [63]. In case of nanocomposites, at least one dimension of the filler is in the nanometre range ($<100 \text{ nm}$) [64,65]. The nanocomposite fillers can be classified regarding their dimensionality. Examples include isodimensional particles, such as nanosilica, one-dimensional cylindrical shapes, such as carbon nanotubes, and two-dimensional fillers, such as layered silicates and graphene sheets [66]. Nanocomposites can also be divided into intercalated (the polymer chains are intercalated between the host layers) and delaminated (the host layers are dispersed in a continuous polymer matrix) forms. Composites with particles in the micrometre range are termed microcomposites [64]. This chapter describes the important properties that are discussed and how these can be tailored.

2.1. Elasticity

The control of elasticity can be of advantage when culturing different cells on PDMS, as it has been shown that cells react to the mechanical properties of their substrate [67]. For medical applications, variable elasticity can also be conducive. PDMS has an elastic modulus of 1–3 MPa (compared to ~50 GPa of glass, and ~3 GPa of polystyrene) [43]. The elastic modulus can be controlled independently of other material properties including surface roughness, surface energy and the ability to functionalize the surface by protein adsorption and microcontact printing. Elasticity of polymers such as PDMS can be tuned by inclusion of fillers [68]. Sylgard 184® already contains silica fillers in its prepolymers to increase its elastic modulus. Microsized ZnO fillers of tetrapodal structure (which reduces problems of agglomeration) were included in PDMS, rendering it even more hydrophobic (WCA was increased from 115 to 160°) and more rigid (at 50 wt% filling the elastic modulus increased to ~5 MPa) [69]. In situ-generated particles, as opposed to mixture of particles with prepolymers, can increase the elastic modulus by more than an order of magnitude. This has been done for a variety of fillers, such as silica through hydrolysis of organosilicates, titania from titanates, and alumina from aluminates [70,71].

Elasticity of PDMS can also be tuned easily by changing the base to curing agent ratio. In the case of Sylgard 184®, the ratio recommended by the manufacturer is 10:1 ($E = 580$ kPa). A 5:1 mixing ratio ($E = 1000$ kPa) was applied to produce more rigid substrates for cell growth, while a 20:1 mixing ratio ($E = 280$ kPa) decreased rigidity and mouse fibroblast cell viability [62].

A variation of the curing temperature also affects the elasticity. It was shown that curing of Sylgard 184® at temperatures from 25° to 200°, using mixing ratios of 10:1 and 20:1, produced Young's moduli of 1.3–2.97 MPa [6,19]. An elasticity gradient was produced by a curing temperature gradient, producing elasticities from 190 kPa to 3.1 MPa [72].

Importantly, Young's modulus and mechanical strength of spin-coated PDMS exhibit a transition from bulk behaviour to thickness dependent behaviour at ~200 μm thickness, which is attributed to the radial orientation of the polymer chains caused by the spin-coating process of the prepolymer mixtures. Interestingly, also the thermal decomposition temperature is depending on the thickness of the PDMS films (310 °C at 500 μm and 450 °C at 50 μm), which is hypothesised to originate from the enhanced cross-linked networks [73].

Varying elastic properties of PDMS can be achieved by mixing different types of PDMS prepolymers, e.g. of Sylgard 184® and Sylgard 527®, at different ratios. In this case, the elastic modulus E covers the range from 1 kPa (down to 0.1 kPa when Sylgard 527® prepolymers were mixed at a 4:5 instead of a 1:1 ratio) up to 1.72 MPa. An application of this approach is the cultivation of cells that physiologically grow in more (e.g. bone matrix) or less rigid (e.g. soft tissue) extracellular environments (Young's modulus $E \approx 0.1$ –100's of kPa) on PDMS of varying elastic moduli [74].

Tissue cells respond to the rigidity of their surrounding extracellular matrix [67], which ranges physiologically from soft (cerebral tissue, ~1 kPa) to rigid (bone, 10–20 GPa) [72] (see Table 2). For example, muscle cells cultivated on the stiffer PDMS substrates formed longer myotubes, but less cell clusters than on the ones cultivated on the softer PDMS [74].

However there are also exceptions; single human epidermal stem cells cultured on PDMS with 0.1 kPa–2.3 MPa stiffness were unaffected by stiffness regarding cell spreading and differentiation. Elasticity was tuned by varying the base and crosslinker ratio from 10:1 to 100:1. Cells were cultured on collagen I (covalently coupled via the heterobifunctional linker *N*-sulphosuccinimidyl-6-(4'-azido-2'-nitrophenylamino) hexanoate (Sulfo-SANPAH). The

Table 2

Comparison of Values for Young's Modulus of Biological Tissues and PDMS. Modified from [75] and [76].

Material	Young's Modulus	
Tooth enamel	50	GPa
Compact bone	20	GPa
Keratin	3	GPa
Basilar Membrane fibres	1.9	GPa
Collagen	1	GPa
PDMS Sylgard 184® (mixed 1:10)	1.6–3	MPa
Human skin epidermis	140–600	kPa
Human skin dermis	2–80	kPa
Cerebral Tissue	1	kPa
PDMS Sylgard 527®	0.1–1	kPa

viscoelastic properties of PDMS of different stiffness, were obtained by load–time curves from microindentation measurements [77].

A last example here is the elucidation of the underlying mechanisms of hyperplasia of myofibroblasts in asthma. Bronchial fibroblasts were seeded on PDMS substrates of varying elasticity (1, 10, 50 kPa; similar to the histological environment; measured by a rotational rheometer). The PDMS samples were prepared at ratios of crosslinker to base of very low 0.012 to 0.030, which leads to a partial polymerisation only. Differentiation of fibroblasts into myofibroblasts was enhanced with increasing substrate stiffness [78].

Gradients of elasticity were also created by mixing the PDMS prepolymers via syringe pumps with varying mixing ratios. Cured PDMS samples were then exposed to air plasma treatment under strain, which after release produced a continuous gradient of wrinkling wavelength [79].

Super soft PDMS (elasticity ≈ 0.1 kPa) has been used to create microstructures by moulding. There the difficulty is to separate the mould from the template. In order to prevent rupture of the mixture with Sylgard 527® during release to form microchannels, curing with so-called candy moulds (sacrificial template material, replicated from the confectionary industry) that can be dissolved by immersion of the sample in water has been demonstrated as a viable option [80].

A focused ion beam (Ga⁺) was used to produce patterns with variable Young's modulus (200 MPa to 2 GPa) [81].

In conclusion, the elastic modulus of PDMS is easily tuneable in a wide range, which renders it amenable to study the cell-substrate behaviour over a range of substrate elasticities, a topic that merits further research.

2.2. Transparency, spectral characteristics, photothermal effect

PDMS is transparent in the UV-VIS range. Transparency is paramount in applications such as transparent electrodes, e.g. for (elastic) touchscreens, OLEDs, solar cells, and wearable electronic contact lenses [82]. It also allows the observation of biological samples via light microscopy, e.g. in single cell cultures. Transparency can be modulated and turn PDMS into an optical filter, by inclusion of molecular dyes and particles, rendering it amenable for applications such as fluorescence microscopy [83]. For example, in an on-chip fluorescence device, optical long pass-filter Sudan II dye was added to PDMS monomer prior to moulding of the microchip, thereby rendering an additional long-pass filter superfluous, while exhibiting negligible autofluorescence. The transmittance above the cut-on (570 nm) was >80%, OD 4 below 500 nm, the roll-on was 50 nm. These values are close to the standard interference filters used in fluorescence microscopy, except for the roll-on, which is much smaller for interference filters, e.g. 508AELP from Omega Optical, Inc. (>85% transmittance above 515 nm, OD 5 to 6 below 490 nm, roll-on of ~8 nm). When working with samples

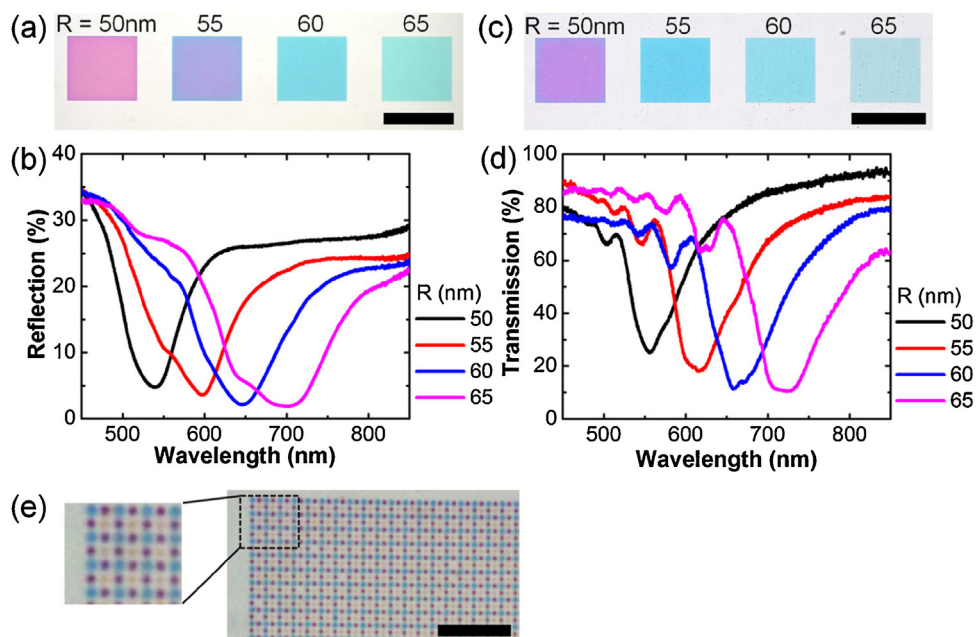


Fig. 3. Optical properties of vertically extruding silicon nanowires embedded in PDMS. (a) Optical microscope images in reflection mode. Scale bar is 100 μm . (b) Measured reflection spectra. (c) Optical microscope image in transmission mode. Scale bar is 100 μm . (d) Measured transmission spectra. (e) Bright-field optical microscope image of PDMS embedded nanowire arrays with different radii. Scale bar is 10 μm [88].

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with multiple fluorescent probes, or normal to small Stokes shift (~ 20 – 25 nm of Alexa Fluor dyes or FITC), a short roll-on yields improved results. However, the cost of the Sudan II based PDMS filter is substantially lower than that of the interference filter (137\$ for the 508AELP filter with 25 mm diameter) [83]. PDMS doping with Sudan II was also applied for PDMS hanging droplet based fluorescence detection using mobile phone cameras, where cured hanging drops of PDMS act as integrated lenses and light filters [84]. Liquid-core waveguides were embedded in dye-doped (Sudan II) PDMS allowing the diffusion of the dye molecules into the waveguides. Dye-doped waveguides exhibited performance comparable to a commercial dielectric filter. However, the sensitivity was limited by the autofluorescence of the dye [85]. A contact image sensor for fluorescence detection of cells was built using a spin-coated PDMS layer doped with Sudan II as emission filter [86]. Filters made from four different commercially available inks (PelikanTM) that were directly mixed with PDMS oligomers were demonstrated for respective transmittances [87]. Dye-free optical filters were constructed by embedding vertical, parallel silicon nanowires in PDMS with differential absorption at visible and near-infrared wavelengths depending on the diameters of the silicon nanowires (radii from 60 to 70 nm, length 1.5 μm). The silicon nanowires extrude vertically from the silicon surface. The colours are not due to scattering or diffractive effects of the collective, but from the guided mode properties of the individual nanowires. See Fig. 3 [88,89].

PDMS doped with colloidal suspension of black micrometre-sized dye particles to provide tuneable opacity to the specimens by mechanical distension was demonstrated [90]. Titanium dioxide microparticle (particle size 270 ± 50 nm) doped PDMS was used to reflect fluorescent signals inside microchannels in order to enhance the collected signal by up to 11-fold [91].

As it is known that the molecular environment significantly impacts important optical properties of organic dyes like absorption and emission spectra, bleaching and dimerization tendency, there is a need for comprehensive and systematic data of organic dyes including fluorophores in a PDMS matrix. However, in contrast to the interference filters typically used in fluorescence applica-

tions, the spectral cut-off characteristics of doped PDMS filters is usually less steep; because of the inherent advantages and cost-efficiency of PDMS, further progress in this regard is highly desirable.

A low cost flexible plasmonic device was produced by deposition of gold nanoparticles (200 nm $\times$ 25 nm) from a silicon wafer surface structured with 300 nm posts to a sinusoidal PDMS surface. The functionalised PDMS layer was irradiated with a polarised light source, and a red-shift of the resonant frequency was observed with increasing strain (up to 30%) [92]. A color-tunable, stretchable luminescent PDMS film was created by inclusion of quantum dots with emission intensity ratios that are dependent on the power of a single wavelength light source. Due to the difficult reproduction of this property, it can be applied as a counterfeit feature [93].

The photothermal effect (plasmonic heating) of metal nanoparticles can generate extremely localised heat. With gold nanoparticles (~ 42 nm diameter) embedded in PDMS (*via in situ* mixing with aqueous AuHCl_4 solution [0.15% w/v]), curing rates at a factor of 4.9 E9 were achieved upon irradiation with 532 nm laser light. Such optically cured polymers may be amenable for 3D printing applications [94].

2.3. Refractive index

PDMS has a refractive index of ~ 1.4 at wavelengths from 589 to 1554 nm [22]. Transparent ZrO_2 -PDMS nanocomposites with high refractive index were created by dispersing siloxane-derivatised ZrO_2 nanoparticles in a PDMS matrix. The refractive index of the resulting nanocomposite could be varied from 1.39 to 1.65 by increasing the ZrO_2 content from 0 to 20.8% v/v. The transparency in the UV-vis domain was $>93\%$ [95]. Likewise, TiO_2 nanoparticle dispersions in PDMS were used to achieve optical materials with a refractive index up to 1.9. A key for success for use of such hybrid materials as optical elements is a strategy for achieving a homogenous dispersion of the particles before and during polymerization despite potentially limited chemical miscibility. This can often be achieved by the use of an alkoxy silane linker that

facilitates interfacing between the siloxane and an anorganic or organic filler [96]. Refractive index tuning allows construction of optical waveguides, solely based on PDMS, or in conjunction with low refractive index plastics like fluoropolymers, and potentially integrated in a microfluidic system. The rapid progress in miniaturized optical systems demanding for new solutions renders PDMS with problem-tailored refractive indices a prime candidate for this application field.

2.4. Electrical conductivity

Electrical conductivity is a requirement for electrical signal propagation in functional materials for applications such as pumps, sensors, and actuators, especially at the microscale. Recent efforts in research were directed towards the realization of stretchable transparent electrodes for applications ranging from microsystems to robotics and wearable electronics [97,98]. PDMS offers suitable material properties, i.e. it is stretchable and transparent; however, it is non-conducting (volume resistivity $2.9 \times 10^{14} \text{ ohm}\cdot\text{cm}$). Traditionally, indium tin oxide (ITO) is used as a material for fabricating transparent electrodes. However, ITO is not stretchable and comparatively expensive [22]. Patterning metallic structures is popular in microelectronics, but the metals do not adhere strongly to PDMS due to the weak adhesion between the metal and PDMS caused by the low surface energy of the latter. Therefore, to improve electrical conductivity, the integration of conducting structures (fillers) into bulk PDMS has been a topic of research: Nanocomposites have recently been under investigation to this end. The nanocomposites used to increase electrical conductivity can be divided into carbon particles (carbon black, carbon nanotubes [99], graphene sheets [99]) and metal particles (e.g. silver particles [100], nanowires, and flakes [101], gold nanotubes [102]). Conductive PDMS, in particular graphene foam/PDMS composites, have also been proposed as flexible electromagnetic interference shielding which can surpass the best values of metals and other carbon-based composites [103].

Comprehensive reviews on stretchable conductors based on nanomaterial composites [104] and on metallic nanowire based transparent electrodes [105] were published recently. A recent review presents the scientific efforts made to achieve flexible electronic devices, inspired by nature [106].

Electrochemical sensor for real-time monitoring of tissues and cells (both in stretched and un-stretched states) was developed by synthesizing gold nanotube networks through galvanic replacement of sacrificial silver nanowires (AgNWs), which had been deposited onto polydopamine treated PDMS. Polydopamine (PDA) is formed by simple immersion of any surface into a solution of dopamine via oxidation by oxygen in the solution and can be used for the immobilization of biomolecules. The optical transmittance at 550 nm was 81.7% (85.4% for AgNWs), the sheet resistance of 23.1 Ohm/sq (18.9 for AgNWs). The proof of principle was done by using it as a sensor for nitric oxide (NO) via cyclic voltammetry with a detection limit of 3 mM NO. Finally, the sensor was inserted into a human umbilical vein to detect NO [102].

An alternating electrical field was applied to cancer cells in microculture, which reduced their proliferation. This was done by fabricated electrodes in a 3D microfluidic setup and by filling microchannels with a PDMS/silver flake (10 μm) mixture. This approach could pave the way to innovative clinical cancer therapies [101].

If the concentration of the added particles reaches a level above the percolation threshold, the conductivity of the composite changes rapidly over several orders of magnitude due to the formation of an interconnected particle network, whereas below the threshold, conductivity relies mainly on tunnelling effects [107,108]. The percolation threshold is mainly governed by the aspect ratio of the filler, where high aspect ratio enables the for-

mation of conductive networks at lower concentrations [109]. For example, percolation thresholds for carbon black are in the range from 1 to 10%, for MWCNTs 1.5–6%, for silver powder and flakes 33–83% [110]. Carbon nanotubes are of considerable interest with regard to both reinforcement and increases in electrical conductivity [111,112]. For MWCNTs the percolation limit is <5% (the sheet resistance decreased from $E+08$ to $E+02 \text{ }\Omega/\text{sq}$) [113]. The disadvantage is the opaqueness and (depending on the application) decrease of conductivity with strain.

The resistivity of nanocomposites can change with applied mechanical strain. Thus, a strain gauge can be manufactured (see e.g. [114]) that can be expressed by the gauge factor (or strain factor), which is the ratio of relative change in electrical resistance R , to the mechanical strain ϵ . For other applications, such as electrodes and conductors, strain should not change the conductivity. Increasing conductivity by increasing conductive fillers is limited by mechanical stability: The concentration of the particles cannot increase above a certain limit, where the composite becomes increasingly brittle and prone to rupture. Another approach, in contrast to mixing with the prepolymers, is the generation of aerogels that are backfilled with PDMS. This has the advantage that contacts between nanoparticles already form contact between each other before PDMS polymerization comes into play. For example, SWCNT aerogels, where the nanotubes are connected to each other via van der Waals interactions, were backfilled with PDMS. This composite was transparent at 30 μm sheet thickness with conductivities of $70\text{--}108 \text{ S m}^{-1}$ at $8\text{--}10 \text{ mg mL}^{-1}$ and an electrical resistance increase of only 14% at 100% tensile stress [115]. In a similar approach, graphene foams were produced by chemical vapour deposition and were infiltrated with PDMS. This composite featured an electrical conductivity of 10 S cm^{-1} at low 0.5 wt% [116].

Graphene foam was infiltrated with PDMS. It was found that larger graphene flakes ($>100 \mu\text{m}$) yielded a higher conductivity ($\sim 3.2 \text{ S/m}$ at 0.4 wt%) as compared to small flakes ($2\text{--}15 \mu\text{m}$; $1.4\text{E-}2 \text{ S/m}$ at 1.9 wt%) [117]. A graphene/MWCNT aerogel was backfilled with PDMS, producing an elastic, stretchable (20% over 100 cycles), electrically conductive composite (2.8 S cm^{-1} , 1.3 wt%) [118].

A three-dimensional graphene aerogel PDMS composite was produced with high electrical and thermal conductivities (1 S cm^{-1} $0.68 \text{ W m}^{-1} \text{ K}^{-1}$) [119]. A carbonized nano-sponge filled with PDMS was applied as strain sensor (gauge factor up to 18) for motions of joints, blood pulse, and breathing [120]. A graphene AgNW nanocomposite was presented as a stretchable conductor, in which a graphene aerogel was filled with AgNW to increase the contact resistance between the graphene sheets. The AgNWs were irradiated with a xenon flash lamp to optically weld them. The resulting PDMS nanocomposite achieved low gauge factor at 60% stretching [121].

A nickel foam was dipped in a graphene oxide solution, the nickel scaffold was etched away and PDMS was infused under vacuum. For a compressive strain of 30% (perpendicular to the electrical current), the resistance of this nanocomposite changed to $\sim 120\%$ of its original value. Application as a blood pulse meter was demonstrated [122].

As graphene is a novel material with very particular characteristics [123] and has been explored only recently, we expect progress using this material in conjunction with PDMS for a variety specific high performance applications, for example, a transparent flexible and stretchable humidity sensor applying a three layer WS_2 (wolframdisulfide, corresponding to 2.1 nm) on PDMS [124].

A wearable strain sensor was achieved by backfilling a conductive carbon sponge with PDMS. The carbon sponge was simply made by pyrolyzation of waste paper and had an apparent density of $\sim 10 \text{ mg/cm}^3$ with a moderate electrical conductivity of 13.6 S/m . The gauge factor was 1.78 until 25% of stretch. The feasibility as a pulse and motion sensor, and as a heater was demonstrated. This

is a promising easily upscalable approach utilizing recycling of a natural abundant material, without inclusion of CNTs or graphene, which have to be fabricated at smaller scale and may pose contamination and health issues [125].

Bacterial cellulose (BC) [126] is an interesting recent approach to nanocomposites of PDMS. It has long been used as the raw material of *nata-de-coco*, an indigenous dessert food of the Philippines. BC can be produced at an industrial scale by simple bacterial fermentation of coconut water. It is composed of interconnected nanofibers, that can be converted into a cellulose aerogel by freeze-drying and into a network of carbon nanofibers (pyrolyzed BC, p-BC) by pyrolyzation under argon at 600–1450 °C. SEM imaging revealed highly interconnected fibres of 10–20 nm diameter. P-BC aerogels are highly flexible. They were infiltrated with PDMS resin and cured at 70 °C. The conductivity of the PDMS/p-BC composite increased strongly by pyrolyzation at >800 °C, probably due to higher degree of graphitization. It reaches up to 0.41 S cm⁻¹ at pyrolyzation at 1450 °C, which is higher than graphene or carbon nanotubes composites of PDMS, due to the networked structure compared to the dispersion where conductivity is governed by percolation threshold. The resistance increase upon stretching is 14.8% by a maximal stretching of 80% [127]. See Fig. 4.

In comparison to other conducting polymers, the mechanical flexibility, the high working temperature range and the chemical robustness of this material renders PDMS based conductive materials highly attractive. Robustly connecting metallic electrodes to conductive, flexible PDMS is still a practical challenge and thus, novel, facile production strategies for such connections could trigger an even broader application range.

2.5. Energy harvesting

Biomechanical energy harvesters are a very active and interesting field of research [128–130] and is only briefly mentioned here. The main concept is a transformation of body movements via stretching of flexible pads to produce electricity, e.g. for wearable electronics. For example, a triboelectric nanogenerator (TENG) was developed, and demonstrated to being capable of continuously powering an electronic watch. This approach is easy to fabricate and producible at low cost [131]. Also, an elastic piezoelectric was produced via embedding of copolymer nanoparticles in PDMS matrices. Upon lateral stretching of up to 600% it dissipates electricity. The piezoelectric coefficient was comparable to commercially available rigid piezoelectrics [132].

A stretchable, transparent nanocomposite nanogenerator for self-powered physiological monitoring (joint angles, eye blinking, pronunciation, peripheral artery pulse) for the detection of lateral strains (up to 60%), and compressive forces was presented. Basically, sprayed AgNWs electrodes were covered by a second layer of piezoelectric nanocomposite BaTiO₃ in PDMS, and third layer of pure PDMS. This design also allowed energy-harvesting upon repeated contact and separation [133].

2.6. Heat

In microsystems, transfer of heat to the microsystem in a controlled manner, transfer of heat away from an exothermic reaction taking place inside a microchannel, and determination of the temperature at specific locations inside a microsystem may be required by specific applications. Here, tailoring the thermal properties of PDMS may come into play.

Thermal conductivity of PDMS is low with ~0.15–0.27 W m⁻¹ K⁻¹ (for comparison polystyrene 0.14 W m⁻¹ K⁻¹) [22]. However, in settings such as microsystems and electronic encapsulation, good dissipation of thermal energy, low thermal expansion, and lightweight are desirable. To

this purpose, thermally conductive fillers are added to the PDMS bulk. Besides the intrinsic thermal conductivity of fillers, the thermal conductivity of PDMS composites depends on characteristics of the filler, such as filler volume fraction (also referred to as loading level), particle shape, particle size, and adhesion between the fillers and matrix, and the thermal property of interface [134].

Carbon-based fillers, such as multi-walled carbon nanotubes, are thermally conductive fillers with high thermal conductivities, high corrosion resistance, and a lower coefficient of thermal over-expansion than metal fillers, such as alumina and copper [63,135]. A list of thermal and electrical conductivities and elasticities of PDMS, PDMS (nano-) composites, PDMS fillers, and the relevant fillers is displayed in Table 3.

Ceramic Al₂O₃ nanoparticles were dispersed in PDMS at 2 and 10% w/w, thereby increasing the heat flow from 0.152 to 0.294 and 0.364 W m⁻¹ K⁻¹ respectively [136].

An optical heater was achieved by chemically functionalizing graphene (CFG) that was dispersed in a PDMS matrix. CFG was prepared by reduction of graphene oxide with dodecylamine, which leads to a good dispersion rate due to the long chain. At 0.05 wt% a temperature increase was measured at ~1 min when irradiated by a 650 nm laser [162]. A microfluidic microheater and temperature sensor was established by inclusion of silver coated copper flakes in PDMS [163].

Closed loop temperature control, and rapid switching of temperatures to prespecified values inside a microfluidic system are still awaiting simple and inexpensive solutions, as this capability is a key requirement for bioanalytics.

2.7. Magnetic

Magnetism in microfluidic designs was reviewed recently [164]. Magnetic forces can be readily employed for actuation of pumps and valves at a microscale without the necessity of wiring from the macro- to the microscale. A micro-pump was created by actuating a PDMS membrane containing coated iron oxide nanoparticles via application of an external magnetic field. The nanoparticles had a diameter of 10 nm and were coated with hydrophobic layers, which reduced their agglomeration [165]. In a similar fashion, PDMS was doped with NdFeB (Neodymium, Iron, Boron; neodymium magnet; currently the strongest permanent magnet) nanoparticles. The composite was characterized and may be amenable to produce microrobots, which could eventually transport cargo such as stem cells [166]. Others added weight fractions of 60–90% of carbonyl iron particles to PDMS prepolymers to create a magnetorheological elastomer, in which the elasticity can be modulated by application of an external magnetic field [167]. A magnetic conveyor belt consisting of an array of 19 mushroom-like actuators (PDMS mixed with carbonyl iron particles) achieved magnetic propagation of super-paramagnetic bead conglomerates with speeds up to 1 mm/s. However, a proof-of-concept for the integration in a microfluidic setup needs to be done [168].

As mechanical actuation of microfluidic systems with precise flow control is still a challenge in many applications, magnetic actuators having the key advantage of not requiring direct mechanical or electrical connectivity to the microsystem, are particularly promising. For further reading see e.g. the recent review about magnetism in microfluidic design [140]).

3. Tailoring surface properties

3.1. Inducing reactive groups

Treatment with oxygen, argon, or air plasma (the most commonly used methods for modification of PDMS surfaces)

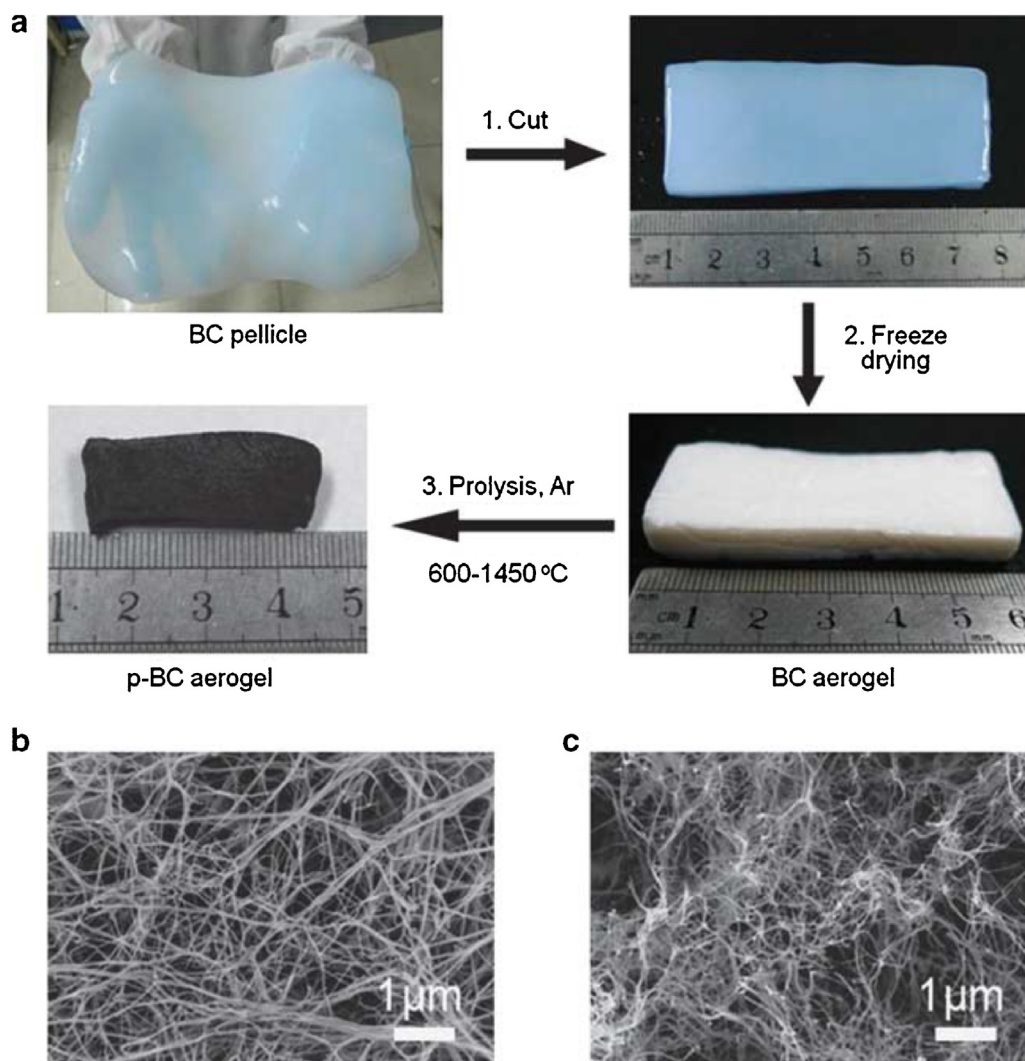


Fig. 4. (a) The fabrication process of BC-based materials. (1) A large-sized BC pellicle ($200 \times 230 \times 3 \text{ mm}^3$) was cut into the desired shape with a sharp blade. (2) After freeze-drying of the wet BC pellicles, the ultralight BC aerogel was obtained. (3) Further pyrolysis treatment of the BC aerogel under flowing argon produced the black p-BC aerogel. (b, c) Scanning electron microscopy images of the BC and p-BC aerogels, respectively [127]. Copyright 2012. Reproduced with permission from the Nature Publishing Group.

[28,169,170], corona discharge or ultraviolet light [171,172] of the PDMS surface destroys methyl groups ($\text{Si}-\text{CH}_3$) and can introduce silanol ($\text{Si}-\text{OH}$) terminal groups and/or generates a layer of silica (SiO_x , mostly SiO_2) [7]. A brittle surface layer of 5 to 30 nm thickness, often containing small cracks is formed, with a WCA of down to $<5^\circ$. This process is self-limited because the diffusive depth penetration of reactive oxygen species is limited by their lifetime [173]. The UV- and UV-ozone-based modification processes are about an order of magnitude slower than the modification by plasma-based treatment. These changes of surface properties were verified by (i) infrared spectroscopy where analysis of the surface of oxygen-plasma treated devices by ATR-IR revealed the presence of a large peak centred at 3200 cm^{-1} suggesting that $-\text{OH}$ groups were present at much higher concentrations compared to untreated PDMS [174], and (ii) XPS measurements, where atomic ratios indicate that UV/ozone treatment removes up to 89% of the carbon from the resultant surface film, leading to an overall stoichiometry close to that of SiO_2 [171,173]. The presence of the silica fillers in the PDMS did not alter the surface properties of the UVO- and UV-modified Sylgard 184[®] specimens [171]. Oxidized surfaces remain hydrophilic when stored in water, however when stored in air they return to a hydrophobic state (termed “hydrophobic recovery”)

within hours due to reorientation of surface hydrophilic groups into the bulk and migration of oligomers from the bulk to the surface, the former process being faster than the latter [7,28,175]. Extraction of oligomers in the bulk prior to plasma treatment with various solvents (e.g. heptane with 5% of weight loss) decreased the speed of hydrophobic recovery [30,169,170]. The hydrophobic recovery process depends also on the thickness of the polymer films, with thicker films ($>1 \text{ mm}$) recovering faster than thin films (in the micrometre range) due to larger reservoirs of free oligomers. The duration of the vacuum prior to plasma treatment and the curing temperature have an influence on the hydrophobic recovery in thin films, due to evaporation of low molecular weight species, especially in thin films. However, comparison of WCA and hydrophobic recovery reported in different publications using plasma treatment of PDMS is impaired due to varying silicone type, sample thickness, vacuum treatment prior to plasma oxidation, curing conditions and plasma conditions [176]. In summary, via plasma and UV/ozone treatment, surface properties such as surface free energy can be tailored and other reactive groups can be introduced easily. The activated PDMS surface can be irreversibly bonded (by covalent binding) to glass or silicon, or surfaces of PDMS, polystyrene or polyethylene treated with air or oxygen plasma. In addition,

Table 3

Thermal and electrical conductivities and resistivities of selected PDMS composites and fillers.

	Filler (w/v)	λ ($\text{W m}^{-1} \text{K}^{-1}$ [298.15 K])	Electr. Resistivity $\text{Ohm}^* \text{m}$ at 298K	El. cond. S/m	E = Youngs mod.
Polymer	PDMS Sylgard 184®	0.27 [22]	2.9E+12 [22]		1–3 MPa [43]*
Composites	PDMS Sylgard 527®				5 kPa [74]
	PDMS Al ₂ O ₃ 10%	0.364 [136]			
	PDMS Ag particles (1–3 μm) 82%		4.35E-5 [110]	2.3E4 [110]	
	PDMS Carbon black (50 nm) 11%		0.11 [110]	9.1 [110]	
	PDMS MWCNT (15 nm* >1 μm) 8%		0.101 [110]	9.9 [110]	
	PDMS Graphene 4%		3.33 [137]	0.3 [137]	
	PDMS Carbon sponge 1%		0.0735 [125]	13.6 [125]	1.1 MPa [125]
	PDMS CNT/graphene oxide 2%		0.0370 [138]	27 [138]	
	PDMS CNT/graphene 1%		0.195 [139]	5.12 [139]	
	PDMS Graphene flakes (0.2%) & gr. Foam 2.7%	1.08 [140]			4 MPa [140]
Fillers	PDMS Graphene foam, 8% carbon black	~0.65 [141]			
	Ag Silver	429 [142]	1.62E-8 [143]		82.7 GPa [144]
	C Graphene (single layer)	2000–5000 [150,151]	2.86E-5–2E-4 [152]		1 TPa [153]
	C MWCNT	~3000 [150]	1E-3 [143]		0.94–0.96 TPa for SWCNT [155]**
	C Carbon Black	6–174 [135]			
	Al ₂ O ₃ Aluminium Oxide	30 [134]	1.2E11 [146]		345–409 GPa [145]

the electroosmotic flow in microfluidic channels with plasma-activated PDMS surfaces is enhanced. Plasma treatment can also be applied inside microchannels [177,178]. Thus, plasma treatment and UV-ozone treatment are simple, rapid and effective methods to induce reactive surfaces, although direct comparison between different studies is not always possible due to the varying technical apparatus employed. Hydrophobic recovery limits the use in temporally extended applications where hydrophilicity is required. The reduction of hydrophobic recovery is discussed in the following chapter.

3.2. Hydrophilic and hydrophobic modifications, anti-fouling

The hydrophobic nature of PDMS limits its applicability for microchip capillary electrophoresis, autonomous microfluidic patterning, and other applications where capillary forces are required. Microchannels with hydrophobic surfaces will not fill with aqueous solutions by capillary forces. Also, proteins and hydrophobic molecules will adsorb passively to the PDMS surface, thereby changing the surface properties and the concentration of analytes in the sample. The inhibition of binding of proteins, bacteria, cells and organisms to a surface (optionally with the formation of a biofilm, which is composed of aggregates of microorganisms residing inside a dynamic polymer network [179]) is called anti-fouling or anti-biofouling. Hydrophilicity and anti-fouling may be desirable surface properties. Anti-fouling coating strategies for PDMS were reviewed recently and are therefore discussed below only in a selective manner [180,181]. Regarding the terminology of the wetting behaviour of surfaces with water, there are 4 regimes: hydrophilic and hydrophobic (10° – 90° and 90° – 150°), and superhydrophilic and superhydrophobic (0° – 10° and 150° – 180°) [182]. Superhydrophobicity is usually achieved by combination of low surface energy and presence of surface roughness [183].

3.2.1. Hydrophilic modifications

PDA coating was used for the immobilisation of BSA to the walls of a microfluidic channel, thereby reducing the WCA from 112° to 41° and improving the electro-osmotic flow. With this approach, enantio-selective chromatographic separation of L- and D-tryptophan was achieved [184].

Hydrophilicity and amino acid repellency was achieved by “grafting onto” of polyamidoamine (PAMAM) dendrimers to PDMS treated with oxygen plasma and silanized with γ -glycidioxypropyltrimethoxysilane (γ -GPS). The WCA was reduced from 108° to 32° , the EOF was reduced allowing the separation of arginine and phenylalanine [185].

Surface modification of PDMS with 3-mercaptopropyl trimethoxysilane (MPTMS) rendered it hydrophilic (WCA 80°). O₂ plasma treatment further reduced the WCA to 64° . Unlike pristine PDMS treated shortly (60 s) with O₂ plasma (WCA 71°), which showed hydrophobic recovery after 1 day, MPTMS functionalised PDMS remained hydrophilic after 9 days by recovering only to values preceding the O₂ plasma treatment. Moreover, reduced crack formation after O₂ plasma treatment was observed, and the elasticity of the surface remained unchanged, opposed to the rigid glass like brittle layer that is formed on the O₂ plasma treated non-functionalised PDMS. This is a simple and inexpensive method to render PDMS hydrophilic and elastic [186].

PEG-divinyle ether was covalently bonded to the surface of a PDMS substrate, that was pre-treated with H₂SO₄, via hydrosilylation reaction producing a hydrophilic surface (WCA 10°) that had the disadvantage of a hydrophobic recovery after two days (WCA 70°). The surface modification increased EOF and reduced adsorption of rhodamine B [187].

Plasma polymerization was used to deposit acrylic acid on PDMS. The surface was activated via Ar plasma treatment followed by an Ar/acrylic acid plasma and then by a He plasma to cross-link the polymer film. Contact angles of water in oil were 80° and decreased to full wetting when sodium dodecyl sulfate (SDS) was added to the water [188].

When the prepolymer is doped with 0.5 wt % undecylenic acid (UDA), the electroosmotic mobility in a modified PDMS channel rises to nearly twice that in the native PDMS channel [189].

Self-polymerization of norepinephrine by oxygen oxidation was used to modify a PDMS microfluidic channel. WCA decreased to 13° . Enantioseparation chiral compounds (D- and L-phenylalanine) was achieved [190].

Hydrophilic modification requires multistep processes in a time-sensitive protocol and there is a need for simplified approaches that will allow single-step hydrophilic modification

with long-lasting stability. Addition of surfactants (Triton-X-100 and PDMS-*b*-PEG block-copolymer) to PDMS prepolymers was presented to reduce the WCA. Reduction of WCA and protein adhesion was observed. However, it is not fully clear how this may affect characteristics of cured PDMS (elasticity, transparency, reactivity) or leads to diffusion of mobile amphiphilic molecules to aqueous solutions in contact with the PDMS surface [191].

3.2.2. Hydrophobic recovery

Plasma treatment renders the PDMS surface hydrophilic. However, as mentioned above, hydrophobic recovery occurs within hours when stored under air. This effect can be abrogated by storage in a polar liquid (e.g. water) via reducing reorientation of hydrophilic groups from the surface to the bulk. Alternatively, it was reduced by oxygen plasma pretreatment followed by exposure to a SiCl_4 plasma and an oxygen– CCl_4 mixture plasma, where the proposed mechanism is reduction of surface coverage by monomers originating from the bulk [192]. The contact angle of PDMS with water can be slightly varied by changing the mixing ratio of oligomer and curing agent [193]. Wetting properties can be tailored via the synthesis of densely grafted copolymers (also termed molecular brushes) which can be accomplished by “grafting onto” (attachment of side chains to the backbone), and “grafting from” (grafting side chains from the backbone), or “grafting through” (homopolymerisation of macromonomers) [194]. Molecular brushes are one-dimensional macromolecules that contain a high density of side chains connected to a linear backbone [195]. Piranha solution ($\text{H}_2\text{SO}_4/\text{H}_2\text{O}_2$) was used to oxidize a PDMS surface to form Si–OH groups, then a surface initiator for atom transfer radical polymerization (ATRP) [196] was immobilized and a polyPEG brush (dendronized PEG-methacrylate) was grafted to the surface via ATRP. This increased wettability (WCA $\sim 33^\circ$) and reduced non-specific protein adsorption by 85%. The WCA increased with time and reached a plateau after 16 h ($\sim 46^\circ$) [197].

The introduction of thiol groups onto a PDMS surface via base-catalyzed equilibration of MTS ($(\text{MeO})_3\text{Si}(\text{CH}_2)_3\text{SH}$) was performed. The thiol-functionalized PDMS surface was modified with hydrophilic molecules containing enes (acrylate and maleic anhydride) via the thiolene click reaction. The anhydride-modified surface could be further modified with small molecules and hydrophilic polymers such as maleic anhydride with further functionalisation with PEG or chitosan producing a stable reduction of WCA over six months. Thiolene chemistry provides a mild and robust technology for activation and functionalisation of PDMS surfaces [198].

In-mould surface modification of PDMS with various ionic and non-ionic surfactants (i.e. PDMS is poured over a dried layer of surfactant and cured) produced a stable (setting/drying cycles) hydrophilic surface after a wet (soaking in water for 20 h) cycle only with PEG-PDMS-PEG (WCA 22° ; 40° after 11 days of dry storage). Other, especially ionic, surfactants (SDS, Tween 20) were washed away from the surface due to weak interactions. It is hypothesised that the PEG-PDMS-PEG layer is entrapped in the surface of the PDMS that is poured and cured on top of it. This seems to be a simple and inexpensive method, not requiring elaborate chemical or physical treatment. However, long term stability of the modified surface should be evaluated [199].

In-mould surface modification of PDMS with various ionic and non-ionic surfactants (i.e. PDMS is poured over a dried layer of surfactant and cured) produced a stable (setting/drying cycles) hydrophilic surface after a wet (soaking in water for 20 h) cycle only with PEG-PDMS-PEG (WCA 22° ; 40° after 11 days of dry storage). Other, especially ionic, surfactants (SDS, Tween 20) were washed away from the surface due to weak interactions. A rapid (30 min), simple approach applied plasma activation of a PDMS surface followed by spincoating with commercial glue (11 μm thickness),

which was cured under UV and heat. The WCA initially was $\sim 14^\circ$, after six months under air it decreased to $\sim 24^\circ$ [200].

3.2.3. Hydrophobic modifications

“Mechanically assembled monolayers” (MAMs) have been developed (as a variety of self-assembled monolayers [SAMs]) to form hydrophobic (WCA $\sim 130^\circ$) surfaces with semi-fluorinated molecules. Basically, mechanically stretched PDMS was treated with UV/ozone and subsequently exposed to semifluorinated trichlorosilane. Release of the stretching rendered the monolayer denser packed than without stretching thereby abrogating the intercalation of polar molecules such as water (Fig. 5). The WCA decreased minimally when stored in water for 7 days (WCA $\sim 125^\circ$). This method can be applied in anti-fouling [201].

For superhydrophobicity surfaces, a low surface energy together with a micro- or nanostructured surface texture, which entraps a thin layer of air between the solid surface and the liquid, is required. The susceptibility of the surface texture to mechanical damage has been a major impediment [202].

A strategy to achieve self-cleaning transparent, mechanically stable surfaces was presented by a superhydrophobic, superoleophobic (‘superamphiphobic’) modification of PDMS. A contact angle of 158° for water and 156° for hexane was achieved by coating with SiO_2 nanoparticles, which were modified by fluorosilanes [203]. The latter two works demonstrate nicely how joint exploitation of PDMS elasticity, PDMS surface modification, and characteristics of nanoparticles can lead to breakthrough results. Such surfaces with extreme properties like the combination of superhydrophobicity and superoleophobicity surfaces have a large promise to appear in products for healthcare where noncontaminatable surfaces are highly desirable, in addition to the numerous consumer and industry applications that call for such properties.

Perfluorinated PDMS was created via covalent attachment of perfluoroalkane-triethoxysilane of differing chain length to the plasma-oxidized surface. The zeta-potential decreased from -77 to -101 mV and the WCA was 111° . This could lead to improved electro-osmotic flow in PDMS microchannels which can be applied as liquid propagation method [204].

In an effort to reduce costs and increase robustness of fluorosilane modifications of PDMS, a durable, superhydrophobic coating of PDMS with the flame-retardant $\text{ZnSn}(\text{OH})_6$ particles was fabricated by pressing the particles into the PDMS solution and then curing it. The WCA was 158° and stable for 8 months. Paper pressed with the PDMS@ $\text{ZnSn}(\text{OH})_6$ showed flame retardancy [205].

UV/Ozone treatment can also be used to induce regular surface structures (multi-generation wrinkling) when treating under stretch with subsequent release. This effect was used to separate particles of different sizes (10 μm , 3 μm , and 67 nm) by segregation in a microfluidic device [206].

A two-step photografting polymerization procedure (photoinitiator absorption and UV irradiation) was used to modify a PDMS surface with a hydrophilic monomer (acrylic acid), thereby rendering the surface wetting properties tuneable from hydrophobic to hydrophilic. The corresponding WCA was lowest after UV exposure above 300 s (WCA = 44°) [207].

Corona discharge inside microchannels generated silanol groups at desired locations. This was followed by sol-gel process [208] with silica forming a solid glass-like silica surface with increased hydrophilicity (76° , stable over 6 days). This layer also inhibited non-specific absorption of non-polar fluorescent molecule BODIPY and the organic solvent xylene. In contrast, the contact angle after corona discharge without further modification increased from $<5^\circ$ to $>80^\circ$ after 2 days [209]. A chitosan/DNA layer-by-layer coating of PDMS microchannels was applied (WCA = 56°), which resulted in pH-dependent electro-osmotic flow, thereby

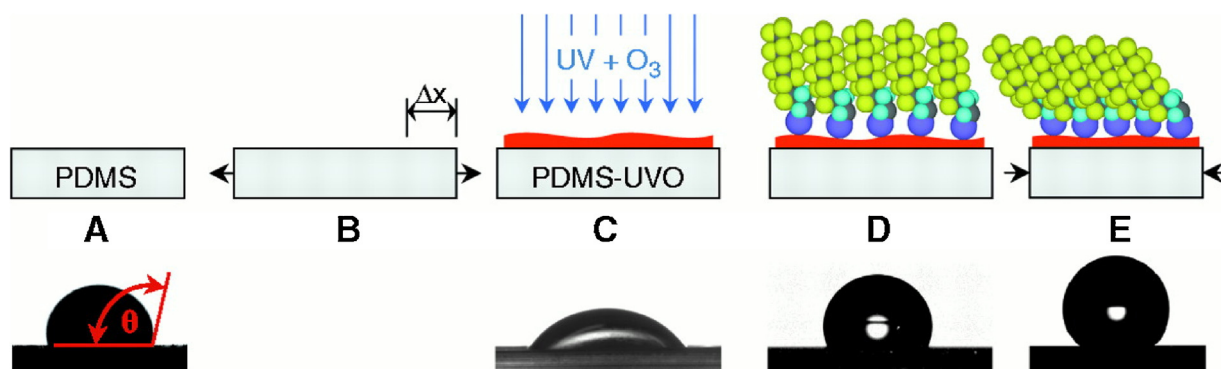


Fig. 5. (A) Pristine PDMS film is prepared (B) After soxhlet extraction in chloroform for 24 h, which removes any non-cross-linked PDMS oligomers, the film is mechanically stretched. (C) Subsequent exposure to a UVO treatment produces hydrophilic PDMS surfaces (PDMS-UVO) (D) The semifluorinated trichlorosilane molecules are deposited from vapor onto this stretched substrate and form an organized SAM. (E) Finally, the strain is released, causing the grafted molecules to form a densely organized mechanically assembled monolayer [201].

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allowing a microcapillary electrophoretic separation of uric acid and ascorbic acid [210].

A composite of PDMS and PTFE particles was cured and the subjugated to (50:50 O₂:CF₄) plasma treatment, which etched PDMS and excavated PTFE particles. This created superhydrophobic microchannels with a maximal WCA of 159° at 15 wt% PTFE [211].

Such hybrid PDMS / fluoropolymers materials may thus combine hydrophobic with lipophobic properties due to their low surface energy and in contrast to the relatively sticky touch-and-feel of native PDMS, such materials may display reduced friction.

3.2.4. Anti-fouling and protective coatings

Anti-fouling (also termed repellency) is desirable in i) medical devices, where the avoidance of plasma protein adhesion to and colonization of implants or catheters with bacteria in a biofilm (that protects them from antibiotics and immune reactions) is pivotal, ii) in contact lenses, and iii) in marine applications. In the latter, the reduction of colonization with bacteria, algae, and molluscs is a goal, thereby leading to i) reduced fuel consumption (up to 40% [212]) in transportation due to increased friction drag ii) avoidance of biocorrosion, and iii) abrogation of transport of invasive species.

An special challenge in marine applications is the achievement of anti-fouling in static conditions. Traditional, ecologically harmful biocidal surface coatings (mostly tributyltin [TBT], which has been globally banned in 2008) are being replaced by functional non-toxic coatings, to meet these multi-billion dollar world-wide requirements [181,213]. Infused PDMS (termed iPDMS) was produced by overnight immersion in silicone oil (kinematic viscosity 5 cSt), that permeates the cured PDMS and forms a slippery surface layer, as confirmed by low water contact angle hysteresis. Medical (reduced colonization by bacteria [214]) and marine (reduced adhesion of mussels) [215] application was demonstrated. It is a facile approach, however the release of silicone oil into the surrounding medium, thereby depleting it from the bulk PDMS, was not addressed.

An anti-fouling PDMS surface with micropillars was created by mould cast, subjugated to treatment with air plasma, immersed with H₂O₂ to introduce silanol groups, which were then reacted with 3-(Trimethoxysilyl)propyl methacrylate. This surface was immersed in PEG-diacrylate, containing a photopolymerizer, which after UV-treatment resulted in a hydrophilic hydrogel (WCA 55°). The WCA remained stable during 30 days (when stored in water; correspondence with the author). BSA-FITC adsorption was reduced (>90%) [216].

A simple and rapid bulk-modification method based on adding an amphiphilic copolymer during the fabrication process was

employed to modify PDMS microchips. Protein adhesion reduction and reduced electro-osmotic flow (EOF), desirable in capillary electrophoresis, could be achieved, when Poly(lactic acid)-poly(ethylene glycol) (PLA-PEG) was used as the additive substance. Compared to the native PDMS microchips, both the contact angle and the EOF of the bulk-modified PDMS microchips decreased [8,217].

Using piranha solution, Si—OH groups were introduced into the PDMS surface, which were then “grafted onto” with quaternized poly(dimethylaminoethyl methacrylate). This created a cationic polymer “brush-modified” surface, with lower protein adsorption (>70% reduction for BSA) and bacterial adhesion (>97% reduction). The WCA was initially ~47° and increased to ~67° after 40 h at ambient conditions. However, the use of piranha solution can be hazardous [218].

As stated above, the formation of biofilms by bacteria is a major issue in clinical settings. They are principally generated by staphylococci via production of exopolysaccharides, proteins, teichoic acids, and extracellular DNA. Besides peripheral venous catheters and urinary catheters, medical implants can be affected by biofilm-associated infections and include: central venous catheters, heart valves, coronary stents, implantable neurological stimulators, arthro-protheses, fracture-fixation devices, breast implants, cochlear implants, intraocular lenses, and dental implants. Biofilm formation can be divided into four steps: (1) initial attachment of bacterial cells, (2) cell aggregation and accumulation in multiple cell layers, (3) biofilm maturation and (4) detachment of cells from the biofilm into a planktonic state to initiate a new cycle of biofilm formation elsewhere [219]. The initial attachment of staphylococcal cell is strongly influenced by the degree of hydrophobicity of the cell surface and that of the matching biomaterial surface.

Coating PDMS with nanoparticles (hyaluronic acid / N-myristoyl-lysine-methyl ester) with 100–250 nm diameter resulted in WCA of 50° (stable under air and water for 60 days). Adhesion of bacteria (*E. coli*, *S. aureus*, *P. aeruginosa*, *P. mirabilis*) and fungi (*C. albicans* [yeast]) was reduced up to 85% [220].

To address this issue of biofilm formation on medical implants, biodegradable, low-toxicity polysaccharide carboxymethyl chitosan nanoparticles were coated onto PDMS. This resulted in a protein (BSA and fibrinogen) repellent, hydrophilic surface. However, the direct inhibition of biofilm formation by adhesion of bacteria was not reported [221].

In a more recent approach, antibacterial and antibiofilm effect was achieved for three antibacterial coatings on PDMS catheter surface. The three coating formulations that were investigated each contained uncharged poly(ethylene glycol) dimethacrylate

(PEGDMA) as cross-linker and a cationic monomer of containing (3-acrylamidopropyl) trimethylammonium chloride (AMPTMA) or oligomeric quaternized polyethylenimine methacrylate (Q-PEI-MA). In an *in vivo* mouse catheter model, a reduction of clinically relevant, often nosocomial, pathogenic bacteria, was 98.88% for methicillin-resistant *Staphylococcus aureus* and of 94.51% for vancomycin-resistant *Enterococcus faecalis* (VRE). The strength of this approach are the *in vivo* experiments, reflecting urinary catheter infections with formation of biofilms [222].

A photobactericidal surface, activated by normal hospital lighting conditions, reduced bacterial growth by $> \log 4$ of *S. aureus* in 1 h and *E. coli* in 4 h. This was achieved by a combination of embedded ZnO particles and a “swell-encapsulation-shrink” inclusion of crystal violet, which are supposed to act via radical and singlet oxygen formation [223].

An anti-fouling, hydrophilic (WCA 20° up to 5000 h of storage at laboratory ambient conditions), biocompatible PDMS surface was created by covalent silanization of zwitterionic sulfobetaine silane. The repelling rates for *P. aeruginosa* and *S. epidermidis* reached 99.86% and 99.78%, respectively [224].

Self-cleaning (transparent) surfaces have great potential in marine anti-fouling and building constructions. The temporal stability of such surfaces should be evaluated in the future.

PDMS also has applications in protective coatings, such as anticorrosion, anti-icing, flame-resistance and anti-reflection, as reviewed recently [225].

3.2.5. Cell culture

As mentioned in the introduction, cell culture in PDMS may differ from cell culture in polystyrene. This can be due to the adsorption and absorption of lipophilic compounds, e.g. anti-cancer molecules, such as tamoxifen, where a simple coating of the PDMS walls with parylene (the bottom consisted of glass) in wells with dimensions similar to 96-well plates abrogated the change of the tamoxifen concentration [36].

PDMS was shown to absorb retinoic acid from cell culture medium, thereby leading to reduced expression of CD38 on CD34⁺ haematopoietic progenitor cells after seven days of culture in wells with PDMS bottoms [226].

A microfluidic platform for the chemosensitivity and resistance assay (CSRA) with cells of patients can guide clinical treatment with anti-cancer molecules. The microfluidic design would allow small-scale co-culture of different cell phenotypes, thus mimicking the *in vivo* 3D pathophysiological processes *in vitro*. Microfluidic devices made of the thermoset PDMS and the two thermoplastics polystyrene and cyclo-olefin polymer, with a multiple myeloma cell line were compared. For PDMS, a $\sim 4.3 \times$ increase (compared to the two thermoplastics) of IC₅₀ (half-maximal inhibitory concentration) of anti-cancer molecules bortezomib and carfilzomib for multiple myeloma was observed. ToF-SIMS data revealed the absorption depths of the two molecules, with only ~ 100 to 300 nm for the thermosets and stronger with ~ 3.5 to $5 \mu\text{m}$ for PDMS after 24 h incubation under a sessile drop. Here, obviously, the permeability of PDMS returns false values that are critical for guided clinical therapy. This can lead to the excess of the upper limit of the therapeutic window potentially resulting in toxic concentrations in patients [227].

The standard substrate for human pluripotent stem cells (hPSC) is a biological matrix (Matrigel®) that suffers from standardization and stability issues. This limitation was overcome by generation of a reproducible substrate for cell culture, especially stem cell culture (which are more susceptible and challenging to cultivate). The authors used PDA, which is formed by simple immersion of any surface into a solution of dopamine via oxidation by oxygen in the solution and can be used for the immobilization of biomolecules. After PDA coating, a second and third layer of carboxymethyl chi-

tosan (CMC) and extracellular matrix peptide vitronectin (VN) was added by simple immersion in water buffer (CMC) and by NHS chemistry (VN), respectively. The WCA of the PDA-CMC-VN surface was reduced from 111° (native PDMS) to 32° . Its amenability as a substrate for the reprogramming of somatic cells into human induced pluripotent stem cells, their culture, and their differentiation into cardiomyocytes and neural cells was demonstrated. This may open avenues in fundamental research and (stem) cell therapies [228]. In a similar approach, PDMS was coated with PDA and gelatine, resulting in increased adhesion, proliferation, and differentiation of embryonic stem cells that were cultured in a microfluidic setup. The behaviour of the cells was also influenced by the mixing ratio of prepolymers (5:1–40:1) [229]. PDA coating of PDMS was also applied for the culture of bone marrow stromal cells, promoting adhesion, proliferation, and multipotency [230]. Similarly, PDA-collagen hydrogel complex was synthesised and coated onto PDMS, resulting in increased surface wetting and improved cell viability and reduced apoptosis (L929 fibroblasts and tendon stem cells) [231].

Surface modification with extra-cellular matrix protein poly-L-lysine via (3-aminopropyl)triethoxysilane (APTES) activation [232] of PDMS surface decreased the WCA ($\sim 40^\circ$), increased protein retention in the solution, and improved growth and neurite lengths of P1 rat cortical primary cells [233].

Folic acid was chemisorbed onto a PDMS-Silane-NH layer via NHS-chemistry. This (together with a nanostructure on the PDMS surface) produced a hydrophilic surface (WCA $\approx 20^\circ$), thereby improving the growth of tumour cells (which have a strong expression of folate receptors) [234].

Surfaces of variable stiffness (base to curing ratio 63:1, stiff regions produced by embedding a photoresist with a thin layer of PDMS on top) and a cross pattern of physisorbed fibronectin on top was cultured with 3T3 fibroblast cells. The cells showed a dependency of spreading in relation to the elasticity of the PDMS substrate [235].

via (3-aminopropyl)triethoxy silane (APTES) and cross-linker glutaraldehyde (GA) chemistry, fibronectin and collagen type I was immobilized on the PDMS surface, rendering it hydrophilic (WCA $< 70^\circ$). Cell culture on these surfaces with mesenchymal stem cells showed increased adhesion and spreading compared with unmodified or fibronectin coated PDMS [236]. Similarly, APTES was used to anchor collagen inside microchannels, resulting in improved adhesion under shear stress as compared to collagen that was physically adsorbed via O₂ plasma treatment. Also, increased adhesion of fibroblasts in collagen type I coated microchannels was reported [237]. In a versatile approach, 3D calcite heterostructures were used to create flexible inorganic artificial scales (also as an exoskeleton when densely packed), served as focal adhesion points for rat skin tissue samples, and to render a flexible SiNW field effect transistor (FET) [238]. As part of a cardiac microphysiological system (MPS), PDMS microfilaments ($60 \mu\text{m}$ thick) were printed onto a strain gauge, serving as the attaching layer for rat cardiomyocytes (cardiac muscle cells) [239].

Water contact angles and anti-fouling properties of pristine PDMS and selected modified PDMS are listed in Table 4.

3.2.6. E-skin

E-skin (electronic skin) is a bionic construct, which is trying to imitate some properties of physiological human skin, such as a stretchability (up to 30%), durable physical barrier, sensing (pressure [also termed touch], strain, vibration, temperature [5 to 48°C], thermal and mechanical damage [pain]), self-healing, dissipation of thermal energy (increased via evaporation of secreted water and increased blood flow), sensing of air flow (via actuation of hairs) [240]. E-skin can be applied in prosthetic skin [241] and robotics. In prosthetics, the main challenges are the transmission of afferent

Table 4

Hydrophilic and hydrophobic modifications of PDMS, their recovery and applications.

Modification	WCA (& over time)	Anti-fouling (what, %), (mechanism)	Application	Ref.
Pristine PDMS	108°–117°			[185,218,228]
O ₂ or air plasma	<15° (58° after 2 d)			[224]
Corona discharge	<5° (>80° after 2 d)			[209]
UV (185 & 254 nm under N ₂)	90° (after 1 h irradiation)			[171]
UV/ozone (185 & 254 nm)	<10° (after 20 min irradiation)			[171]
polyPEG brush	33° (plateau 46° after 16 h)	85% protein	Adhesion barrier, antifouling, func. Mat. Med.microdev.	[197]
Quaternized poly(dimethylaminoethyl methacrylate)	47° (plateau 67° after 40 h)	>70% BSA; >97% bacteria	Antibacterial, Functional surfaces for biomed. Microdev.	[218]
UV-cured glue (GL-03®)	14° (24° after 6 months under air)		DNA quantification	[200]
3-mercaptopropyl trimethoxysilane (MPTMS)	80° (stable after 9 d in air)		Flexible electronics, cell culture	[186]
semifluorinated trichlorosilane	130° (125° after 7 d in water)		Super-hydrophobic barriers	[201]
Fluorinated SiO ₂ nanoparticles	158° (156° hexane)		Transparent self-cleaning surfaces	[203]
PDMS@ZnSn(OH) ₆	158° (stable over 8 months)		Self-cleaning surfaces	[205]
Acrylic acid	Tunable down to 44°		Control of wetting	[207]
LBL chitosan/DNA	95° (2 wk in H ₂ O at 4 °C)		Electrophoretic separation of uric acid and ascorbic acid	[210]
PDA-CMC-VN (polydopamine-carboxymethyl chitosan-vitronectin)	32°		Stem cell culture	[228]
PDA-BSA (PDA only)	41° (73°)		Enantio-selective separation of L- and D-tryptophane	[184]
PLL (Poly-L-lysine via APTES)	40°		Cortical cell culture	[233]
PAMAM (polyamidoamine) dendrimers	32°		Electrophoretic amino acid separation (Arg, Phe)	[185]
PEG-PDMS-PEG	22° (40° after 11 d in air)		Self-driven microfluidic channels	[199]
PEG	10° (70° after 2 d)		μTAS	[187]
Hydrophilic hydrogel	55° (stable 30 d in H ₂ O)	>90% BSA-FITC	Biomed. microdevices	[216]
Folic acid	20°		Tumour cell culture	[234]
Sulfobetaine silane	20° (stable >5000 h at lab cond.)	>98% BSA, ~100% bacteria, no lipid adsorption	<i>In vivo</i> implants	[224]
PTFE particles (15%)	159°		Contact free dosage of liquids	[211]
PEG & chitosan (via click chemistry)	71–73° (86–92° after 6 months in air)			[198]
Hyaluronic acid/N-myristoyl-lysine methyl ester	50° (stable 60 days in water and air)	~100% BSA, 85% bacteria and fungus	Medical catheters	[220]

signals from e-skin into appropriate regions of the brain (interfacing – also *via* peripheral nerves) translated into cortical neural code (which is still not understood completely) with high signal density. Touch sensing can be achieved by various methods, such as resistive, capacitive, piezoresistive, optical, magnetic among others. Usually, touch sensors consist of a pattern of sensors to allow signal localisation. A simple, low cost approach for touch sensing based on single-layered piezoresistive MWCNT-PDMS composite films was presented, where machine learning (deep neural network, DNN) was applied to measure pressure and position simultaneously from electrodes mounted to the margin of the film. This precludes complicated patterning, however requires DNN training for every such device, currently impeding large-scale fabrication [242]. E-skin is an exciting topic that is investigated very actively. A multitude of potential applications awaits, however integration of all functionalities of human skin remains a key challenge. In this regard, PDMS

is suited to serve as an ideal substrate due to its easy processing, stretchability and biocompatibility.

3.2.7. Self-healing

Self-healing (also termed self-repairing) polymers are a recent focus of attention, especially in the context with electronic skin (e-skin) and protective layers (from light, heat, mechanical damage, and oxidation). In view of recent reviews on this topic [243,241] we limit this chapter to a short introduction. Self-healing polymers can be divided into two categories: i) composites with healants or healing agents (in microcapsules or tubes), also termed *extrinsic* self-healing; and ii) polymers with dynamic, reversible, covalent and non-covalent bonds (such as hydrogen bonds or the Diels-Alder reaction), also termed *intrinsic* self-healing. Extrinsic self-healing cannot heal more than once at a specific location. Extrinsic self-healing of PDMS can be elicited for instance by redox potential [244] or exposure to sunlight [245]. In 2012, the siloxane equili-

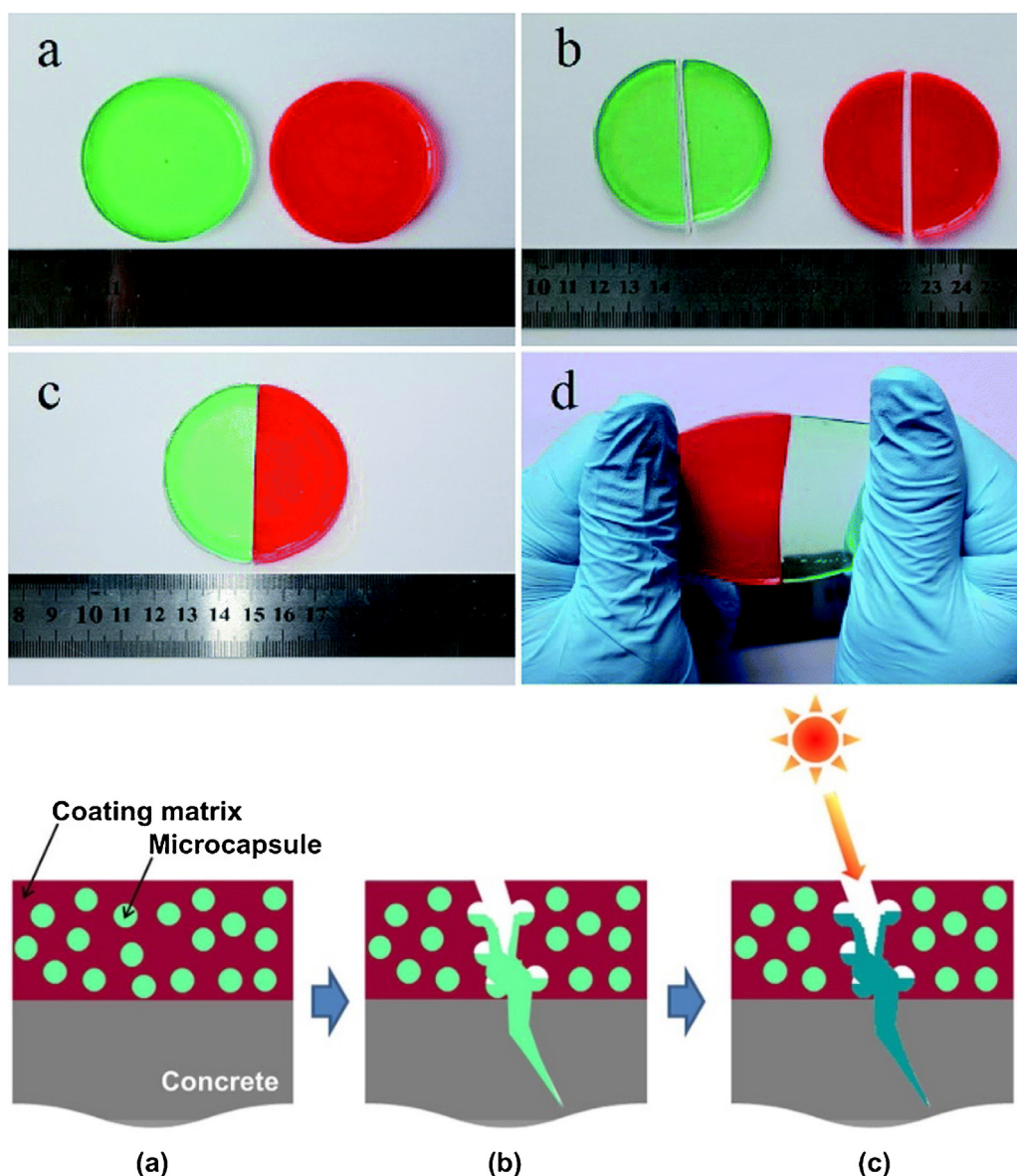


Fig. 6. Intrinsically self-healing PDMS-polyurea. Top: Damaged samples are merged together, leading to self-healing within 48 h at room temperature [247]. Copyright 2017. Reproduced with permission from the Royal Society of Chemistry. Bottom: Mechanism of extrinsic self-healing upon exposure to sunlight [245]. Copyright 2013. Reproduced with permission from the American Chemical Society.

bration (living silicone rubber, living reactive anionic species) that was discovered in 1954, was rediscovered for intrinsically self-healing applications. A mixture of octamethylcyclotetrasiloxanes (D-4) and bis(heptamethylcyclotetrasiloxanyl)ethane (bis-D₄) is heated to 90 °C for 24 h in order to achieve self-healing [246]. Another approach used a composite of PDMS-polyurea with an organic anti-foulant with rapid environmental degradation (4,5-dichloro-2-*n*-octyl-4-isothiazolin-3-one, DCOIT) and was tested successfully during six months for intrinsic self-healing at room temperature and marine anti-biofouling. A limitation is the gradual release of the anti-foulant, which limits the service time of the coating [247]. (see Fig. 6) A superhydrophobic nanofilament (nanograss) surface based on PDMS, infused with silicone oil and trichloropropylsilane was presented. It achieved complete recovery of superhydrophobicity within 1–2 h at room temperature after chemical and physical damage [248].

3.2.8. Soft robotics

Soft robotics aims to re-engineer properties of soft sensors and actuators found in nature, such as senses and muscles in vari-

ous species with (e.g. ray) or without (e.g. jellyfish, octopus) rigid structures (skeleton). A photo-tactic artificial ray consisting of a gold skeleton, rat cardiac muscle cells, and a matrix of PDMS with varying mixing ratios showed fascinating behaviour, mimicking motions of real rays. Three layers of PDMS mixtures (Sylgard 184 and Sylgard 527) were (spin) cast on top of each other resulting in varying elastic moduli (see Fig. 7) [249]. Likewise, a completely soft, microfluidic octobot made of PDMS with the ability of actuating its tentacles *via* decomposition of H₂O₂ was created [250].

3.2.9. Dry adhesives

Biological dry adhesives can be found in the adhesive pads on the digits of geckos [251]. They increase adhesion and friction, allowing the clinging to vertical and inverted, hydrophilic and hydrophobic, rough and smooth surfaces, and, although adhesion being strong, attachment is temporary, with the ability to detach and reattach rapidly. This is achieved *via* hierarchical micro- and nano-structures. Several gecko inspired dry adhesives based on PDMS have been developed in the past decade, only for dry [252] and a dry and wet adhesive utilizing mussel adhesion (based

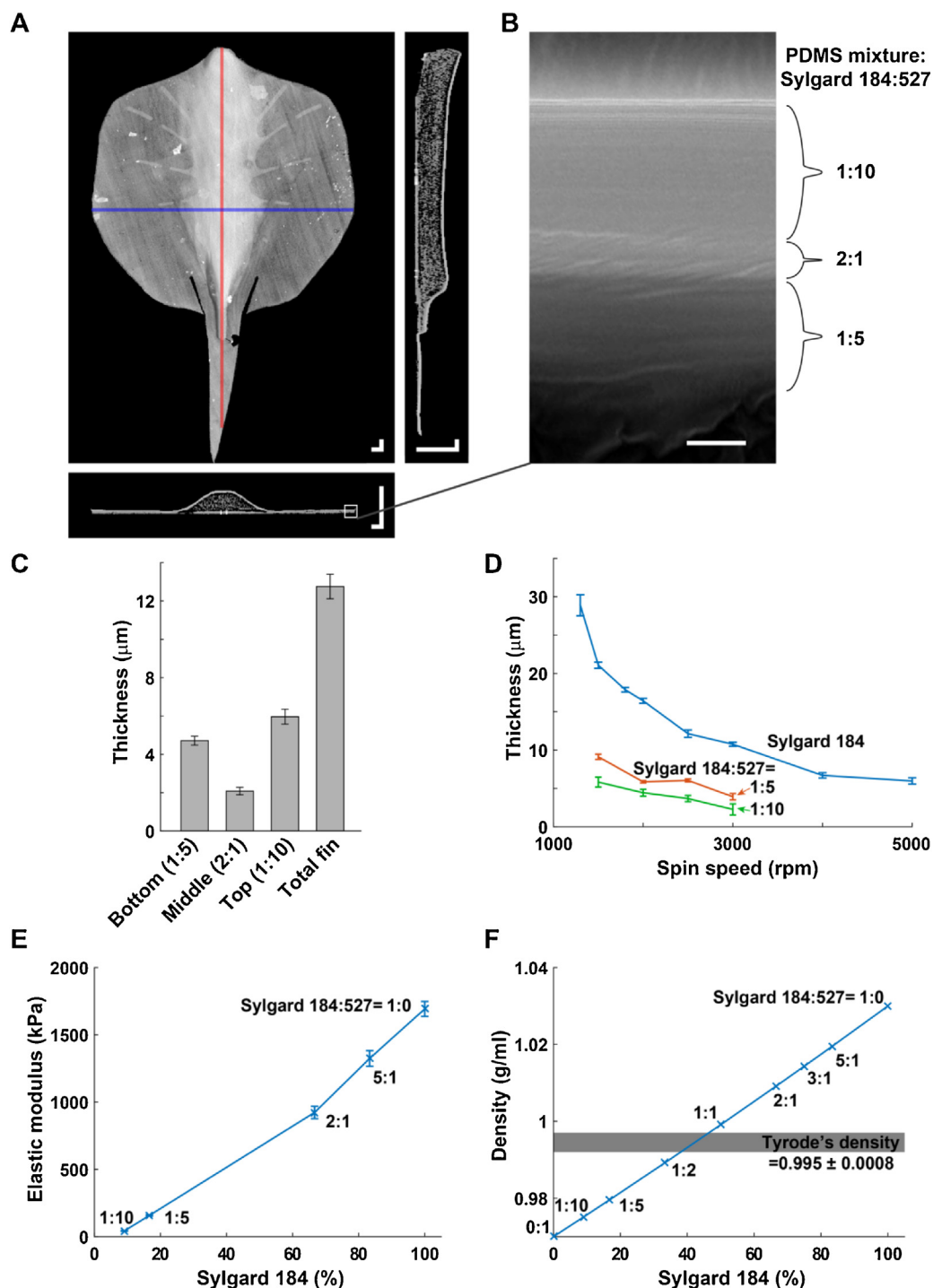


Fig. 7. (A) Micro-CT scans of the tissue-engineered, soft robotic ray. (B) SEM (scanning electron microscopy) image of the three layers of PDMS (scale bar 2 μm). (C) Thickness of three PDMS layers. (D) Thickness of three films of PDMS mixtures upon varying spinning speed. (E and F) Elastic modulus and physical density of various mixing ratios of Sylgard 184 and Sylgard 527 [249].

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on 3,4-dihydroxy-L-phenylalanine) inspired polymers also for wet conditions [253]. Adhesives for wet conditions as used by tree frogs were investigated more recently. One example reproduced the hierarchical structures found on digits of tree frogs with PDMS micropillars with embedded polystyrene nanopillars. The authors claim their approach is simpler than gecko-inspired adhesives with their complicated fiber geometries [254]. Another example created

nanosucker modified PDMS, mimicking octopus suckers operating at wet or dry conditions. However, the attachment duration is limited by diffusion of air into the nanosucker cavities. Also, attachment-release for more than 30 cycles deforms the nanosuckers [255]. Such bio-inspired adhesives can exceed performance of biological adhesives and may be used in medicine (electrodes, re-attachable wound healing patches), bioelectronics, biomechanics,

and robotics, where adhesion and friction control are highly desirable.

3.3. Coatings

Coating of PDMS with surfactants (molecules with a hydrophilic and a hydrophobic part) has been used widely to reduce fouling of surfaces and increase wettability of its surfaces. This is due to the fact that the hydrophobic part will adsorb to the PDMS surface, while the hydrophilic part is oriented towards the liquid phase in microchannels. The use of coatings to modify PDMS surface chemistry in order to reduce fouling was reviewed recently [180].

Physisorption methods change surface properties conveniently without chemical reactions (e.g. covalent bonding) involved. The mechanisms involved are polar-polar interaction, hydrogen bonding, and van der Waals forces. A disadvantage of physisorption methods consists in potentially weak (electrostatic, hydrophobic) interactions of the coating agents with the surface, which may lead to desorption and thus result in instable coatings. In general, there are two strategies to apply physisorbed coatings: static and dynamic. In the first case, the coating is incubated for a specific time and then rinsed with pure buffer removing unbound residues. For dynamic coatings, the coating reagent is present in all solutions during an experiment. Therefore, if the coating substance partially desorbs, it may be replaced with coating molecules present in the used solution, thus overcoming some disadvantages of physisorption methods in case of weak interactions. Physisorption of proteins to PDMS can be employed to prevent adsorption of analyte proteins in the analyte solution and to increase hydrophilicity. Bovine serum albumin is used routinely as such a biopassivator (or blocking agent) [10,256]. Covering with BSA reduces the WCA [184]. Coating PDMS with extracellular matrix proteins, such as fibronectin or collagen is employed to ameliorate the attachment of cells in culture [13].

Another approach is the use of zwitterionic molecules (i.e. molecules with positively and negatively charged moieties but neutral over-all charge). A zwitterionic triblock copolymer poly(carboxybetaine methacrylate)-poly(propylene oxide)-poly(carboxybetaine methacrylate) (PCB-PPO-PCB) was applied onto a PDMS surface via “graft onto” hydrophobic-hydrophobic interaction, which reduced protein and cell attachment. The reduction of non-specific binding of proteins (fibrinogen and lysozyme) and blood plasma was confirmed by surface plasmon resonance (SPR). This design also allows the introduction of functional groups for biomolecule interaction (e.g. binding to amine groups of proteins such as antibodies via NHS chemistry). In order to demonstrate the ability to bind biomolecules in an assay, an antibody was covalently attached to the zwitterionic blocks via NHS chemistry, with subsequent capture of a biomolecule from solution with high sensitivity [257]. A superhydrophilic zwitterionic material, poly(carboxybetaine methacrylate) (pCBMA) was introduced to the PDMS surface via surface initiated atom transfer radical polymerization (SI-ATRP) with a trichlorosilane initiator. Polymerization of CBMA occurred in 100% water, without the need of organic solvents (which would diffuse into the PDMS bulk). Zwitterionic polymers are strongly hydrated and have extremely poor solubility in nonaqueous solvents, making surface rearrangement with PDMS thermodynamically unfavourable. Also, zwitterionic polymers have strong chain-chain interactions formed by ionic bridging, which improves the stability of the coating. Film thickness was measured by ellipsometry, with a resulting measured thickness of 30 nm. These films rendered the surface hydrophilic (WCA not reported) and abrogated PDMS hydrophobic recovery. The surface also becomes repellent to protein adsorption, as demonstrated by detection of fibrinogen adsorption by an anti-fibrinogen-ELISA showing ~2% adsorp-

tion compared to untreated PDMS. The pCBMA film is repellent over 74 days, when stored under wet conditions (31 days when stored in air). It can be functionalised via covalent attachment of biomolecules containing primary amine groups, e.g. antibodies or RGD-containing peptides for bioassays. To this purpose, the surface was activated, i.e. the exposed carboxylic acid groups reacted with reactive *N*-hydroxysuccinimide (NHS) esters using 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC) [258].

By varying the mixing ratio of the Sylgard 184® prepolymers (1:1–1:10 curing agent to base v/v), some Si-H groups are preserved even after curing. Via hydrosilylation reaction, allyl carboxybetaine (ACB) was “grafted onto” these groups on the PDMS surface (PDMS-CB). The WCA was reduced from 103° to 46° and was stable after 3 months stored in dry air. The PDMS-CB surface was demonstrated to be biocompatible and anti-fouling (reduction of BSA adsorption of 75%, and bacterial adhesion reduction >95%) [259].

An antifouling method using engraftment of PEG was created by a microfluidic device constructed of PDMS and SU-8. The PDMS surface was functionalized with amines by treatment with air plasma activation and reaction with 3-aminopropyltrimethoxysilane (APTMS). The SU-8 surfaces were amino functionalized using ethylene diamine (EDA). This was followed by modification of both PDMS and SU-8 surfaces with mPEG-NHS, thus rendering the surfaces hydrophilic (contact angle with water of PEG-grafted SU-8 surface ~55° and of PEG-grafted PDMS ~64°) and protein repellent (64.5% reduction in nonspecific adsorption of BSA). However, the grafted PEG layers on the SU-8 surfaces were found to be stable only for 5 days [260].

Hematocompatibility is desirable in surfaces of implanted biomedical device that are in contact with body fluids. PDMS coated with a polydopamine/ hyaluronic acid composite reduced the adhesion of platelets and the inflammation markers (e.g. TNF- α) of adherent macrophages, making it amenable to implants with improved biocompatibility compared to unmodified PDMS [261].

PDMS treated with oxygen plasma was modified with three different coatings, namely Pluronic™ (POE-PPO-POE triblock copolymers F_{108}), poly(l-lysine)-g-poly(ethylene glycol) (PLL-PEG), and the hybrid coating *n*-dodecyl- β -d-maltoside and methyl cellulose (DDM/MC). Binding to static and dynamic coating were employed. Reduction of EOF was achieved for all coatings. For PLL-PEG, a reversal of EOF was observed, which, in combination with the other coatings would allow to titre the EOF to zero. All coatings were hydrophilic, and reduced the binding of bovine serum albumin by 45–74 %. However, the stability of the coatings was not addressed [262]. Adsorption of the ionic complementary peptide EAR16-II to native and oxidized PDMS surfaces rendered them hydrophobic and protein repellent (>96% reduction of adsorption of BSA and lysozyme). The peptide forms a stable β -sheet amphipathic film. Microchip electrophoresis of Arg-FITC and Phe-FITC was done using EAR16-II as dynamic coating, thereby reducing the adhesion of the analytes [263]. A microfluidic spectroscopic sensor for ascorbic acid measurement via three different immobilization strategies of ascorbate oxidase was presented, including alumina sol-gel encapsulation, direct physisorption and physisorption with alumina xerogel modification, with the least being the best strategy [264]. Thus, there are different strategies to calibrate the EOF of the PDMS surface inside microfluidic channels, with zero EOF required for electrophoretic separation, and high EOF for flow induction. A concomitant goal is the reduction of non-specific binding of analytes (which change the concentration in the microchannels) and matrix molecules (which can change the EOF upon adsorption). Different approaches were presented, however, the temporal stability (due to surface reorientation)

is still an issue for some or and is not sufficiently addressed yet.

3.4. Optical layers

Interfacing a microsystem with the macro world, e.g. for signal readout, is often done by optical means because microscope technology is a well developed field, fluorescence technology is rapidly advancing, fluorescence readout one of the most sensitive signal transduction techniques, and because optics do not require structural access to the microsystem. Tuning the optical behaviour of a microsystem to a specific application by optimizing its material properties is thus a promising strategy for innovative solutions. A tuneable phase grating (with potential application to coherent light beam) was created by a layer of (relatively rigid [~ 3 GPa]) polystyrene (PS), which was transferred from a silicon wafer to a pre-stretched cured PDMS sample by immersion into water. Upon increasing strain release the surface commenced to wrinkle accordingly. When coherent light passed through such a wrinkled surface, a local phase shift was induced with a magnitude proportional to the wrinkling amplitude. Thus, the intensity of the diffraction peaks can be tuned by three orders of magnitude [265]. In order to tailor wettability and optical properties, silver nanoparticles were sputtered onto PDMS consisting of varying ratios of oligomer and curing agent (3:1, 5:1, 10:1, and 20:1). A reduction of contact angles with water from 114° to 98° was achieved. The optical transmittance showed a plasmon resonance spectral shift [266]. A stretchable reflective optical grating was created by moulding PDMS on a grated substrate, metallizing it (different equivalent thicknesses up to 160 nm) by implanting with silver nanoparticles with a supersonic cluster beam implantation apparatus equipped with a pulsed microplasma cluster source. This produced higher fidelity reproduction of submicrometer grooves than the use of thermally evaporated silver. These nanocomposite surfaces were stretchable and resilient at $>10,000$ stretch cycles. They can create optical elements in curvilinear conformal designs [267]. An elastomeric polymer resonant waveguide grating structure was created that can be used as a pressure sensor. The sensor can be used to measure applied surface pressure optically, which has potential for optical ultrasound imaging and pressure wave detection/mapping [268]. Integrated optical devices with multiple functionalities, such as waveguides and diffraction gratings may introduce novel composite microdevices in the near future.

3.5. Electrically conductive layers

Intrinsically conductive polymers (also referred to as 'synthetic metals', e.g. polyanilines and polypyrroles, have a long history, but were limited in the parameter scope they can cover due to poor reaction control and poor processability. However, recently the development of novel polyaniline materials (such as polyaniline nanofibers and polyaniline composites, e.g. with graphene [269]), as well as polypyrrole has opened new avenues to applications such as sensors [270], supercapacitors [271], electrochromics [272] and artificial muscles [273] and thus render it a burgeoning field. On the other hand, novel PDMS composite materials that combine advantages of PDMS with the advantages of metallic, or carbon-based, or polymeric conductors are a hot topic of recent research that addresses the development of flexible (bendable and twistable) and stretchable electronics. These ongoing fields of research will be discussed in the respective chapters. Topics range from lab on a chip designs (sensors and actuators) [274], (bio-)medical devices [275,276], biomechanics, prosthetics [241], soft robotics [249,250], flexible energy storage [277,278] to flexible or stretchable, wearable, conformal electronics (also termed epidermal electronics) [76,106,279–282], wearable sensors for personal

healthcare [283–285], tactile sensors [286], artificial skins (e-skin) [240,241,287], electronic muscles [288], self-healing electronics [240], and 'lab-on-skin' [289]. Approaches can be divided into bulk and surface modifications, and in terms of the conductive material applied. Modifications created include (i) PDMS (nano)composites containing electrically conductive additives such as carbon black, graphene, metal particles, or carbon nanotubes (see chapter above on bulk modifications), and (ii) surface modifications with materials such as tin oxide, silver nanowires [114], gold [290,291], silver nanopatterns [292], silicon [293], multi-walled carbon nanotubes (MWCNTs) [294], single-walled carbon nanotubes (SWCNTs) [295], and graphene sheets [296]. The different approaches yield different conductivities and changes thereof upon stretching, which renders them amenable to different applications. One-dimensional nanomaterials (metal nanowires and carbon nanotubes) for soft electronics were reviewed recently [275].

A challenge is the interfacial adhesion of metal films to PDMS during stretching, due to their different Young's moduli. Freestanding metal films rupture at 1–2 % inelastic stretch, but bonded to an elastic substrate they can be stretched up to 20% without fatigue. This applies for gold if the film is not continuous, i.e. contains nano- or micro-cracks [297]. These phenomena were studied and modelled with polymers other than PDMS [298,299] and are discussed in more detail in the chapter on gold below (3.5.8 Gold).

Depending on the application, the conductivity upon stretching is intended to be reduced (application as a strain sensor) or to be maintained (application as electrodes or conductors). For wearable skin-mountable stretchable electronics a Young's modulus in the same range as the skin and a stretchability of 30% or more are desirable [300]. The data for the Young's modulus of human skin are non-homogeneous, due to the multi-layered architecture of the skin, the strong dependence on the measurement method (e.g. indenter tip diameter), the location on the body, and *ex vivo* or *in vivo* experiment designs [301] (practical values for skin electronics are listed in Table 2).

Generally, for applications not demanding a strain sensor, a stable high conductivity is desirable. A selection of different approaches and their gauge factors can be found in Table 5. Electrically conductive polymers and conductive polymer coatings are cornerstones of the rapidly growing field of flexible electronics that was pioneered about two decades ago [302]. There are various strategies to produce integrated electronic surfaces, i) rigid islands connected by flexible conductive elements; ii) placing the transistor on a pre-stretched PDMS upon release producing a rippled surface; and iii) intrinsically stretchable semiconductors. A fourth method was presented recently, which was based on a mogul pattern (similar to a buckled ski slope) as shown in Fig. 8, created by soft lithography, that allowed stretching biaxially up to 50%. It is simpler in manufacturing than the rigid island approach and is stretchable in two dimensions, opposed to one dimension in the prestretching approaches. This mogul approach achieved excellent stability of electrical resistivity upon stretching of up to 50% [303].

Many devices (touch screens, OLEDs) require electrodes with high transparency and conductivity. The current standard is indium tin oxide film technology ($\sim 90\%$ transparency at 550 nm; low sheet resistance of ~ 20 Ohm/sq). However, these films are brittle and increasingly costly due to scarcity, therefore demanding for alternatives. Carbon based conductors often do not meet the requirements, and noble metals are rather costly. These are the reasons, why CuNWs may be a viable option [304].

An overview of conductivities of stretchable PDMS modifications applying CNT composites, polymers as conductors, and Ag particles was recently published [240], as well as a classification of elastic skin-like electronics that incorporate surface-integrated transistors [241].

Table 5

Stretchability, gauge factors, number of stretch cycles and applications of selected PDMS composites.

Material	Gauge factor	Maximal stretch	Repeatability	Application
Graphene / CNT composite	0.36			[310]
AgNW	2–14	70%		[114]
AgNW networks	10–40	5%		[292]
AgNW	1.1–55.2	0.5–2%		[324]
AgNW	7	80%		[325]
CuNW	0.82	80%		[323]
Au nanomesh		160%	1000 cycles	[326]
Graphene/CNT particles (directional anisotropy)	20'000	100%	1'000	[315]
Graphene oxide	7–173	150%	5'000	[316]
Aligned CNT	4739	0.9%	(10E6 other publications)	[317]
LBL-SWCNT	15	80%		[318]
SWCNTs perpendicular to strain	0.82	280%	10'000 at 150%	[311]
CNT/PDMS foam	15	15%		[319]
Carbon black	5.5–1.8	80%	>30	[320]
Graphite nanobelts	5.1–2.9–0.79	40%	5000 at 2% strain	[321]
Graphite films	522–11344	~50%		[322]
ZnO piezoelectric fine-wires	1250	1%	>60	[313]
electrospun PDMS fiber membrane and polypyrrole (PPy)	0–600 (~10E30)	100% (200%)	>150	[314]
Metal foils	2	5%	1	[312]

For more details, the following reviews are recommended: wearable sensors for health monitoring [289], printing of sensors and electronics over flexible substrates [305], flexible electronics in general [306], carbon nanotube based flexible electronics [307], microfluidic electronics [308], a comparison of carbon nanotubes with graphene [309]. Below is a short selective overview of recently published work.

3.5.1. Electrically semiconductive composites

Intrinsic semiconducting polymers such as PDOT:PSS are increasingly used in applications but the limited mechanical properties of those polymers often render a layered approach on a robust, transparent polymer such as PDMS desirable. A stretchable semiconducting polymer based on 3,6-di(thiophen-2-yl)-2,5-dihydropyrrolo[3,4-c]pyrrole-1,4-dione (DPP) repeating units and non-conjugated 2,6-pyridine dicarboxamide (PDCA) moieties introduced directly in the polymer backbone as hydrogen bonding units was developed. It forms crystal domains and inter-connecting amorphous stretchable areas. The stretchability relies on the presence of hydrogen bonds that can reversibly rearrange. Cracks in the 'skin-inspired stretchable organic transistor' can be healed by thermal and solvent vapour treatment. To qualitatively evaluate the stretching properties of the polymers, thin films are supported on polydimethylsiloxane (PDMS) substrates and stretched to different strain ratios (0%–100% strain, without the formation of microcracks) [327] See Fig. 9.

Interfacing such stretchable materials with conventional, metallic electrodes is an area that still awaits methods that are simple and reproducible to manufacture and result in long-term stable contacts.

3.5.2. Silicon

Scalable aligned arrays of nanoribbons based on high-performance single crystalline silicon complementary metal-oxide semiconductor (Si-CMOS) integrated circuits were produced as a proof-of-principle with the goal of producing stretchable electronics with performance similar to rigid electronic systems. Stretchable transistors, amplifiers, logic gates and oscillators were

demonstrated without compromising the electronic properties upon bending. In short, the silicon array was transferred onto a pre-stretched PDMS substrate with a maximum of 10% pre-strain (above which cracking upon release occurred), upon release producing a nonlinear wavy surface due to buckling, with physics similar to accordion bellows. Positioning of the electronic circuits in the neutral mechanical plane prevented rupture due to bending of the substrate. The Si-CMOS/Polyimide system had a thickness of $\sim 1.7 \mu\text{m}$ (see Fig. 10) [293]. In another approach, transfer of rigid silicon islands connected with thin silicon layers to ozone activated bi-axially pre-stretched PDMS resulted in lifting off of the connecting layers upon stretch-release [329]. This is a highly desirable development because currently, characteristics of intrinsically semiconducting polymers are insufficient for certain demanding electronic applications like high-frequency switching, high current density, and high integration density, that are reachable now with the silicon-on-PDMS approach.

3.5.3. Graphene

Graphene production is generally inexpensive and simple [330]. Graphene consists of a two-dimensional (2D) sheet of covalently bonded carbon atoms, which forms the basis of both 3D graphite and quasi-1D carbon nanotubes. Its intrinsic strength reflected in its Young's modulus of $\sim 1.060 \text{ TPa}$, exceeds that of any other material. Further properties are unusual to other materials and include high room-temperature electron mobility ($2.5 \cdot 10^6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$), very high thermal conductivity ($\sim 3000 \text{ W m}^{-1} \text{ K}^{-1}$), and impermeability to gases. Despite being only one atom thick, graphene is found to absorb a significant (2.3%) fraction of incident white light, a consequence of graphene's unique electronic structure. It can also easily be chemically functionalised [331]. Therefore graphene can be used as heat and electrical conductor and reinforcement in composite materials. In addition, assemblies of graphene with carbon nanotubes creates hybrid materials that have unique advantages compared to single CNT or graphene setups [332]. A graphene layer was transferred onto a pre-stretched PDMS substrate, thereby producing a strain sensor with strains up to 30% [333]. Or, a strain sensor was produced

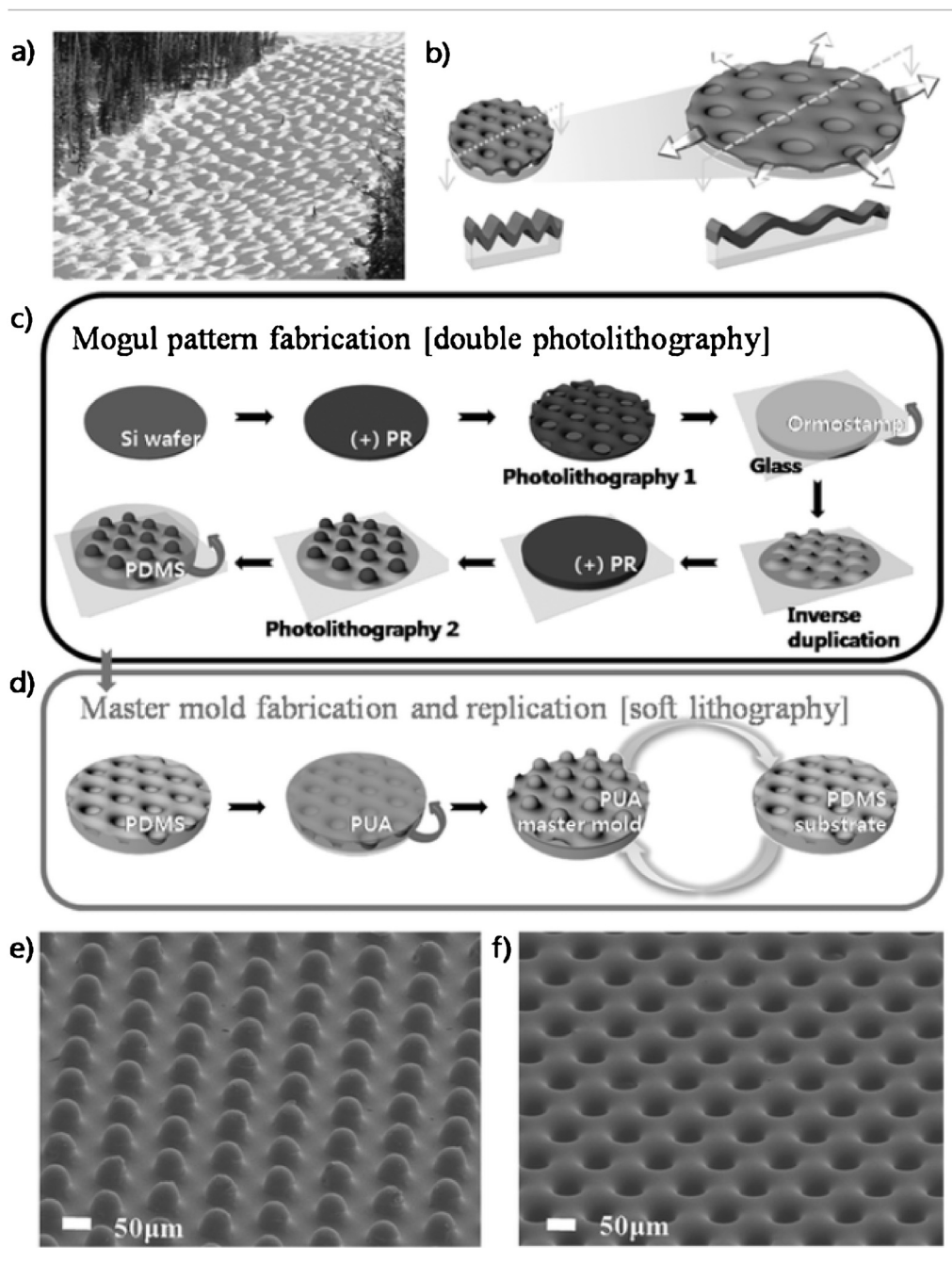


Fig. 8. Stretchable substrate with a mogul pattern and its fabrication process. (a) Bumps and valleys of mogul skiing. (b) Stretching of a mogul-patterned substrate enables efficient absorption of the stress induced on a multidirectional wavy structure. (c) Fabrication process of the mogul pattern on a glass substrate through double photolithography. (d) Replication between PDMS and PUA through soft lithography. Tilted-view FE-SEM images of (e) the PUA master mold and (f) PDMS substrate with the mogul pattern [303].

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by a graphene / CNT composite with a gauge factor of 0.36 with high reproducibility [310]. Chemical vapour deposition (CVD) of graphene layers onto a nickel substrate can be transferred via dry transfer process (etching of the nickel substrate) to surfaces, e.g. a PDMS surface to create conductive and transparent ($\sim 80\%$) graphene films. The films exhibit a very low sheet resistance ($\sim 280 \Omega$ per square) and it was shown that the quality of the graphene layer created by CVD is similar to the one created by mechanical cleaving. This allows the creation of electrodes for flexible electronics. The process of fabrication is simple and easily scalable depending on the size of the chamber of the CVD device

[296]. Ultra-thin graphite layers were deposited on prestrained PDMS creating a buckling surface upon release. Stretchability of 40% was achieved [334]. In a follow-up study, the buckled graphene surface was characterized by AFM and micro-Raman mapping [335].

Graphene woven fabrics were created by CVD (using methane as the source) on a copper mesh that was then etched away. Using this approach, a strain sensor with gauge factors up to 10^6 for a strain of $\sim 8\%$ was achieved, being the highest reported so far for PDMS substrates [336]. A low-cost method to produce electroconductive reduced graphene oxide (RGO) sheets on porous PDMS thin films was presented. It resulted in large 150% maximal strain,

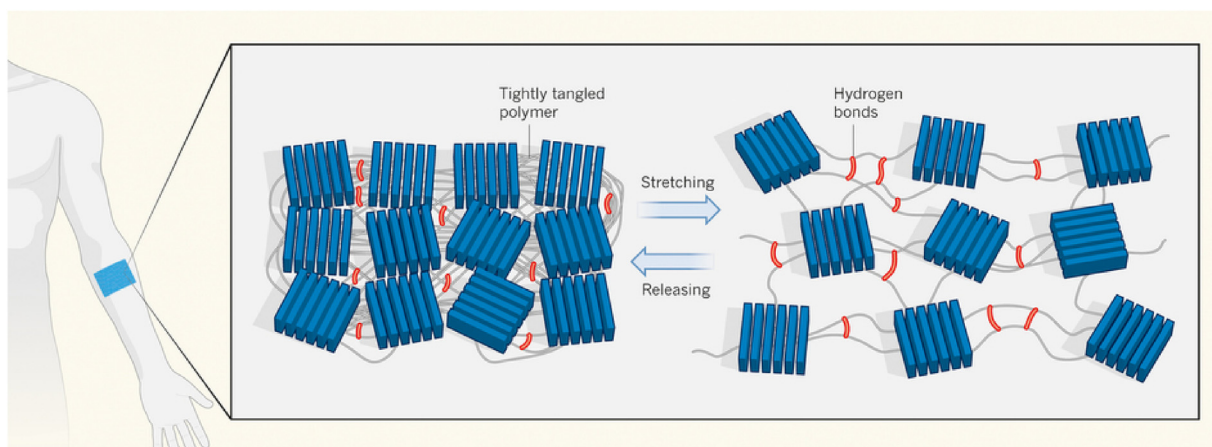


Fig. 9. An intrinsically stretchable polymer semiconductor relying on crystal rigid domains and interconnecting amorphous domains containing hydrogen bonds that rearrange upon stretching. This design may be applied as wearable electronic device or electronic skin [328].

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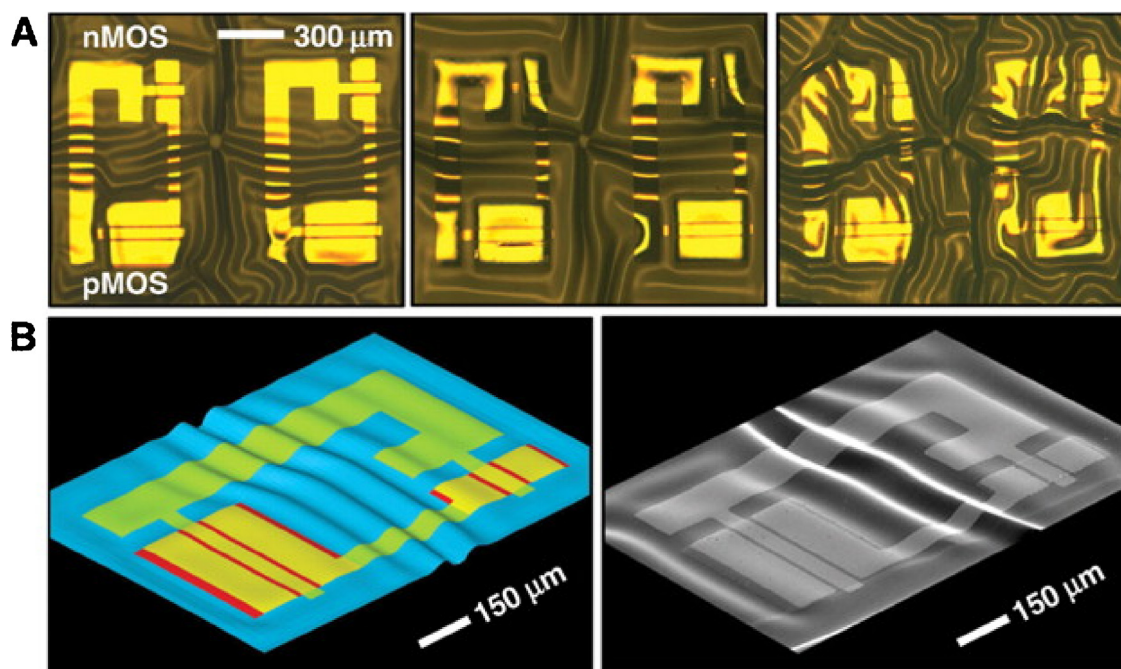


Fig. 10. (a) Wavy Si-CMOS inverters on PDMS, formed with various levels of prestrain. (b) Structural configuration determined by full 3D FEM of a system formed with 3.9% prestrain (left) and perspective scanning electron micrograph of a sample fabricated with a similar condition (right) [293].

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stability over 5000 stretch cycles, a low sheet resistance ($1.5 \text{ k}\Omega$ per square), and a gauge factor of 7–173 with a sensing range of up to 40% strain. Application of this design created dry electrodes (not requiring conductive wet gel) as demonstrated by acquisition of bio-electrical signals (ECG, EEG, and EMG [electrocardiography, electroencephalography, electromyography]) and as a strain sensor. This concept is advertised to function as a low cost device in the internet of things [316].

To conclude, graphene holds great promises for different applications (strain gauge, electrodes) in conjunction with PDMS.

3.5.4. Carbon nanotubes

Carbon nanotubes are becoming increasingly popular to tailor surface properties of PDMS. A strain sensor using aligned carbon nanotubes was built by connection with bundles of CNT between two electrodes [317]. A poly(methyl methacrylate) (PMMA) micro-

machined mould was inked with spin-coated conductive PDMS multi-walled carbon nanotubes (MWCNTs) nanocomposite, which was then microcontact printed on a chlorotrimethylsilane derivatized (to help debond the PDMS from glass) glass surface. PDMS was cast on top of the print, cured, and removed together with the nanocomposite layer. This created a pattern of a conductive nanocomposite of PDMS on top of a layer of pure PDMS. Application thereof is a flexible strain sensor. A relative resistance change of 113% was recorded when the sample was under the highest tested tensile strain of 41.9% [294].

By spray-coating SWCNTs on a PDMS surface transparent stretchable skin-like sensor was produced with strains up to 150% and a conductivity of 2200 S cm^{-1} [295].

Extremely high aspect ratio of super-growth single-walled carbon nanotubes (SWCNTs) were deposited to a film which was coated with PDMS to form an elastic conductor (57 S cm^{-1}) with

a stretchability of 134%. By printing organic transistors onto the PDMS surface, a rubberlike active matrix was formed, which can conform an arbitrary curved surface with a biaxial stretchability of 70% without changes in the transistor characteristics [337]. A layer-by-layer deposition from water was achieved by modifying SWCNTs with conjugated polyelectrodes, thus circumventing the use of organic solvents and sonication to deal with agglomeration due to π - π interactions. 25 layers were deposited on the APTES and HCl treated surface, producing a transparent (75% VIS) conducting layer ($R_s < 5 \text{ Ohm/sq}$), stretchable at 80% with a gauge factor of 15 [318]. This shows the potential of carbon nanotubes; CNT films can lead to the development of flexible displays, flexible field-effect transistors and conducting films as chemical and biological sensors, flexible strain and pressure sensors [307]. For further reading we recommend the recently published review on carbon nanotubes in flexible and stretchable electronics [338]. A microfluidic capacitive sensor was presented. It consists of two parallel microchannels separated by a wall of CNT/PDMS nanocomposite as a dielectric layer, enabling the sensing of pressure, temperature change, and lateral movements of pressure simultaneously. However, a limitation is the unimodal readout that could lead to confounding of the origin of the multimodal signal inputs. The ionic liquid was 1-butyl-3-methylimidazolium tetrafluoroborate ([BMIM][BF₄]). Capacitive changes were monitored by application of an AC voltage (1 V, 1 kHz) [339].

An electronic skin (e-skin; see also chapter 3.2.6) sensor inspired by the human fingertip was demonstrated in a three-layer setup, comprising i) a double coil of CNT-PDMS (4 to 6 wt% CNT, 10 μm long) which acts as a sliding sensor ii) a sandwiched PDMS layer as spacer iii) a base layer of porous PDMS produced via sugar casting, bulk modified with CNTs, serving as pressure sensor. This is intended to reflect three layers of the human skin, which are finger print, epidermis, and dermis. Dermal physiological sensor cells can be divided into fast adapting (FA) and slowly adapting (SA), corresponding to pressure/sustained touch and vibration/dynamic touch sensing. Work has already been published using capacitors and resistors for SA and piezoelectric and triboelectric effects for FA. Exploitation of the triboelectric effect allows more freedom of choice of the material and lower cost than piezoelectric effect. In this approach, pressure is measured by change in piezoresistance of the lower PDMS layer, FA by frequency of alternating voltage of the triboelectric nanogenerator. The combination of sliding and pressure sensing is the outstanding feature of this work, as demonstrated when the sensor is mounted to a glove with capability of sensing the roughness of fabrics, opening avenues for integrated robotic manual sensing [340].

3.5.5. Graphite

Graphite nanobelts (nanostructure in the form of a belt) (10–50 $\mu\text{m} \times 1\text{--}5 \mu\text{m} \times 5\text{--}50 \text{ nm}$), produced by liquid phase chemically assisted exfoliation, were deposited onto PDMS surfaces using the Langmuir-Blodgett technique. The resulting films were stretchable and had a gauge factor of 5.1–0.79 [321].

3.5.6. Silver

While most metal nanoparticles are prone to rapid oxidation, silver is sufficiently noble for production of stable small-sized metallic particles, rendering this material an interesting candidate for hybrid materials at small scale.

Raman spectrum is a powerful tool to study biological macromolecules and has potential in miniaturized settings. Established methods, such as electron beam lithography are being overcome by fast, inexpensive and simple approaches. In a microfluidic setup with AgNP decorated walls a good signal to noise ratio in surface-enhanced Raman scattering (SERS) was achieved. The adsorption of Ag NP (65 nm diameter) to PDMS was promoted by treatment with O₂ plasma followed by APTMS [(3-aminopropyl)trimethoxysilane]

silylation, which introduces amidogen to the surface. The flow in the microchannels conveys sufficient heat dissipation of the impinging laser light. Interestingly, SERS detection of molecules in the microchannel was inversely proportional to flow speed, which was attributed to increased desorption of the molecules with flow [341].

In a different approach, a microfluidic SERS device was built based on *in situ* generated silver nanoparticle nanowalls serving as active SERS substrate [342].

A flexible LED display was achieved by sandwiching a 30–35 μm thick electroluminescent layer (EL), soaked with PDMS, between two layers of PDMS containing transparent AgNW electrodes. By the inclusion of BaTiO₃ particles in the EL, the light intensity was improved by 700% compared to previous work. Their high dielectric constant increases the charge separation in the EL particles, and thus the light emission. The EL particles were 14 fold larger than the BaTiO₃ particles, which allowed filling of the voids of the former's layer. The composite was stretchable biaxially to an area increase of 50%. A flexible copper printed circuit board was bonded to the contacts by stencil printing of silver particle-PDMS (22 vol%) [343].

Ag films (10–20 nm thick) were deposited on pre-stained, UV-treated PDMS, resulting in wavy patterned electrodes with a stretchability of up to 20% without significant resistance change (up to 600 stretch cycles), and a tuneable, optical wavelength dependent optical transmittance. See Fig. 11. Upon applying a voltage, the Ag film turns into a stretchable thin film heater [344].

A layer of nanocomposite of silver nanowires (AgNWs; with an average diameter and length of 150–200 nm and 10–20 μm , respectively) was fabricated by casting AgNWs (synthesized by a modified polyol method) in an isopropyl alcohol solution onto a patterned glass slide and by drying it. This layer was then transferred to and sandwiched between two PDMS layers, which resulted in a flexible, stretchable sensor that changes resistivity upon stretching. A sensor glove for static and motion detection to move an avatar hand could be created based on conductivity changes upon stretching of the sensor. The sensors are stretchable up to 70% with excellent tuneable gauge factors in the ranges of 2–14 and exhibited a good stability, response speed, and repeatability [114]. See Fig. 12.

A stretchable, transparent nanocomposite nanogenerator for self-powered physiological monitoring (joint angles, eye blinking, pronunciation, peripheral artery pulse) for the detection of lateral strains (up to 60%), and compressive forces was presented. Basically, sprayed AgNWs electrodes were covered by a second layer of piezoelectric nanocomposite BaTiO₃ in PDMS, and third layer of pure PDMS. This design also allowed energy-harvesting upon repeated contact and separation [133].

AgNW patterns on PDMS were created by using a poly(*p*-xylylene) (parylene) stencil technique [324]. A microfluidic concentrator based on PDMS/PEDOT:PSS (poly(3,4-ethylenedioxythiophene)-poly(styrenesulfonate)) was developed and applied to the DNA-MO (morpholino) hybridization assay. Interfacing to any microarray platform was facile and rapid [345].

In another investigation, AgNW were deposited on pre-stretched, ozone treated PDMS. Depending on the ozone treatment, upon release, a buckled or a coil structure emerged. The coiled structure displayed a very large stretchability (~104%) up to the fracture of PDMS. This is more than 15 times the intrinsic fracture strain of freestanding AgNWs with the same diameter and about 4 times the stretchability of the in-plane wavy AgNWs. These properties could be employed in stretchable electronics [346].

AgNW (~40 $\mu\text{m} \times 100 \text{ nm}$) networks with various surface coverages were produced by transfer printing from a filter membrane, enabling a strain sensor [292]. AgNWs provide a simple and affordable way to create electrically conducting layers in PDMS. Upon stretching, the electrical resistance changes, thereby producing

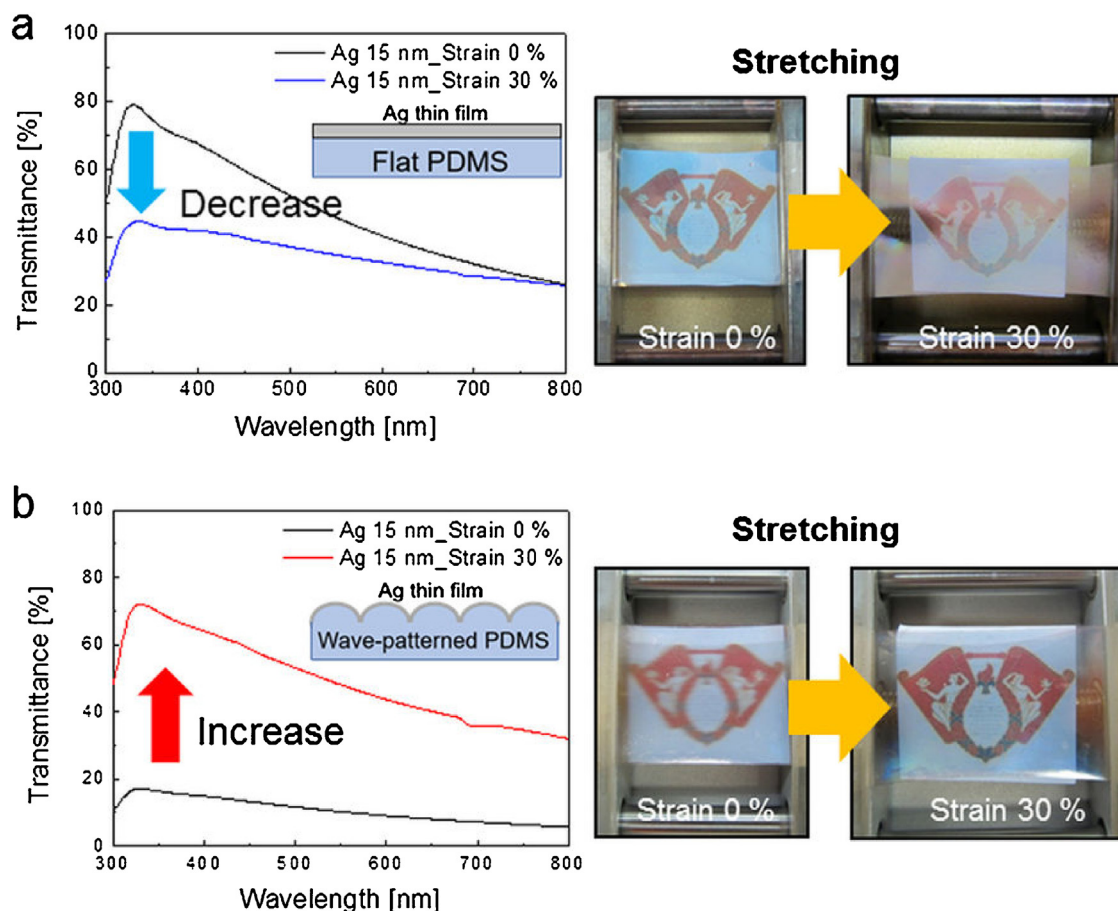


Fig. 11. Optical transmittance of (a) Ag electrodes on flat PDMS substrate and (b) Ag electrodes on wavy PDMS substrate before and after stretching of 30% [344]. Copyright 2017. Reproduced with permission from the Nature Publishing Group.

a strain gauge sensor. This feature was demonstrated in various applications, e.g. a sensor glove, or its use in robotics.

A three-electrode electrochemical lab-on-a-chip glucose sensor was built. The concept relies on inkjet printed silver nanoparticles that are attached to the PDMS substrate via (3-Mercaptopropyl)trimethoxysilane modification [347]. AgNW patterns were fabricated via parylene stencil technique (using lithography and reactive ion etching), by spray coating AgNW in isopropyl alcohol, then the parylene stencil was removed and the AgNW were thermally annealed at 190 °C. A strain sensor and a capacitance based tactile sensor (via through hole PDMS layer sandwiched between two AgNW patterns) were demonstrated [324].

Partially aligned AgNW were welded under solar light exposure at the water/air interface and transferred to PDMS. The resulting composite is stretchable up to 80% strain with full recovery to initial resistivity upon release due to the welded contact. Application as a patterned heater was demonstrated. (see Fig. 7) [325].

3.5.7. Copper

Copper is frequently used in conventional electronics system due to its high electric conductivity, relative inertness and low price. These properties may convey interesting functionalities to microsystems and hybrid materials incorporating PDMS and copper. A problem with high aspect ratio copper nanowires is their propensity to entangle and form clumps, and copper in general to oxidise rapidly (within days). This may be the reasons why there are less publications of PDMS modified with copper than with gold or silver. A capacitance sensor based on stretching of two layers

of CuNW separated by a PDMS layer was built. Partly polymerized PDMS was brought into contact with CuNW on a PTFE filter, leading to partial embedding after removal of the filter. However, upon repeated stretching, the resistance increased with cycle numbers, limiting its function as a strain sensor or electrode [323], thus calling for efforts to render such desirable layers more mechanically stable. In order to overcome problems in interwire electrical contact between CuNWs due to organic compounds and oxide layers, treatment in H-plasma was reported to be an effective, though simple solution [304]. Electronic ink based on CuNW in hydroxypropyl cellulose (HPC) was created with the ability to be painted to surfaces such as PDMS, paper, latex, and rubber, thereby enabling easy on-the-fly printing of electronic circuits and strain sensors [348].

3.5.8. Gold

Gold is highly stable (does not oxidise as compared to Ag and Cu), highly conductive, amenable to nanoparticle and nanorod synthesis and in addition demonstrates fascinating characteristics only apparent when used at the nanoscale, namely its surface plasmon effects rendering it an optical absorbing and sensing material and magnetic properties that are observed in gold nanoparticles. Three different morphologies of gold films on PDMS were described: microcracked, (micro)buckled, and smooth, with their corresponding practical applications (e.g. stretchable electrodes, stretchable interconnects, and dielectric elastomer actuator, respectively). Factors that govern the morphology of gold films on PDMS are deposition temperature, film thickness, elastic modulus of PDMS, adhesion layer thickness, surface properties of PDMS (O_2 plasma treatment), and mechanical prestrain during deposition. Low

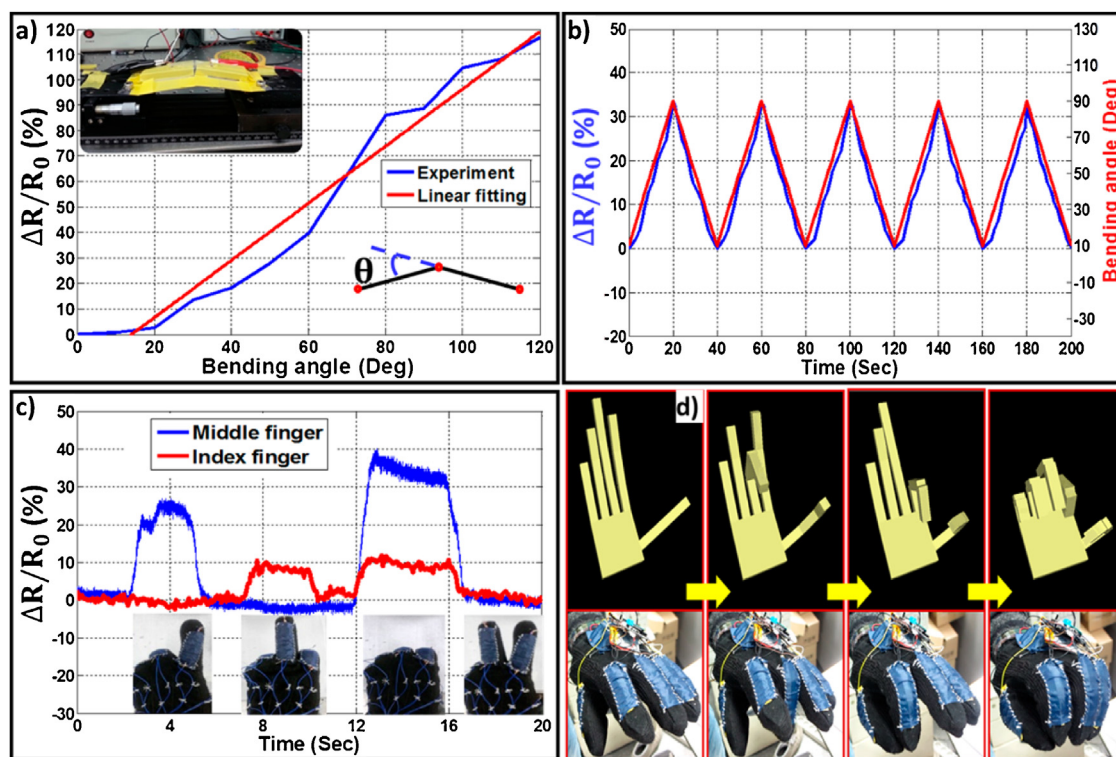


Fig. 12. Human motion detection by the sandwich-structured AgNWs/PDMS nanocomposite strain sensors: (a) response of the sandwich-structured strain sensor to the bending angles from 0 to 120. (inset: photograph of the artificial finger); (b) response of the strain sensor under repeated bending/relaxation cycles (c) motion detection of index and middle fingers; (d) control of avatar fingers in the virtual environment using wireless smart glove system [114].

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adherence of gold layers applied directly to PDMS is a practical problem.

Cr and Ti are good adhesion layers for gold films on PDMS. Cr may be preferred for electronic applications as it can be sputtered and thermally evaporated, whereas electron-beam-evaporated Ti is used in biomedical applications due to its biocompatibility. Ti and Cr are electropositive metals and therefore less mobile on plasma-activated than on untreated PDMS [349].

To increase the adhesion of gold to PDMS, 2–100 nm thick layers of Cr or Ti, followed by 50–100 nm thick layers of gold were sputtered to partially cured PDMS (to avoid cracks due to thermal residual stress) [350] to plasma pretreated PDMS [351], or heated PDMS (resulting in the creation of 20–50 μm buckles upon cooling to room temperature) [352]. Stretchable (up to 22%) surface conductors were created by layers of 100 nm gold on 5 nm chromium via sputtering [290]. Electron beam evaporation of 20–100 nm Au on 3 nm Cr on a pre-stretched PDMS membrane resulted in a stretchable (30%) conductor [291].

A simple and mild method to produce gold films on PDMS via coil-based resistive thermal flash evaporation was reported. It utilises evaporation of metal in vacuum, which is commonly used in microfabrication, production of metallised plastic films (e.g. PET as a barrier to gas and light) and optical mirrors. Vacuum is important to achieve homogenous films. The heating of the gold coil to 1500 K elicits the formation of gold nanoclusters with a diameter of 2–5 nm that impinge onto the PDMS surface without carbonisation thereof. It has advantages compared to gold deposition by e-beam and sputtering onto PDMS, such as generation of smaller cracks and less delamination, indicating stronger adhesion. Furthermore, with sputtering, the specimen may be damaged by high energy atoms (e.g. Ar^+), and with electron beams by x-rays and stray electrons. Conductivity upon 200% stretch was maintained, however the strain gauge and resistivity curve were not communicated [353]. In

another effort to overcome the weak adhesion of gold to PDMS by van der Waals forces, a nanomesh of gold was attached by covalent S-Au bonds via 3-mercaptopropyltrimethoxysilane (MPTMS). Due to poor stretchability, prestretching was applied, which resulted in stretchability of up to 160% with a gauge factor of ~ 1.3 [354].

Bonding of thin Au films onto PDMS was performed by dry peel-off from silicon wafers with (3-mercaptopropyl)trimethoxysilane (MPTMS) as a molecular adhesive (prepolymers are simply poured onto the functionalised gold surface). However, stretching experiments were not conducted [355]. In an alternative approach, the nature inspired surface structuring of gold films with nanopiles (0.5–4 μm length) increased the adhesive strength and stretchability (up to $\sim 40\%$). It was applied as a strain sensor (motion detection) and as an electrode for EMG (electromyogram) [356].

One method to attach interconnects to gold circuits in PDMS is by using a substrate such as glass [357].

Grain boundary lithography was invented to produce a transparent nanomesh of fully interconnected metal wires of gold, avoiding the high junction resistances between nanowires. It has a good electrical conductivity, is transparent, and is stretchable to 160% with modest increase in resistance. Basically, Au was deposited via electron beam evaporation between grains of In on a SiO_x surface. The nanomesh layer was 50 nm thick, with a line-width of ~ 90 nm and a mesh size of ~ 700 nm. The sheet resistance was lower ($\sim 7 \Omega$ per square) as compared to CNT, graphene, commercial ITO films and solution-processed Au nanomeshes. Interestingly, the sheet resistance drops below R_0 until stretched at 80% [326].

Gold nanowire-impregnated tissue paper was sandwiched between two PDMS layers (one blank and one with patterned interdigitated electrode arrays) to produce a pressure sensor capable of sensing blood pulses. The electrode array was produced by Ti/Au deposition with a shadow mask and electric beam evaporator [358]. A microfluidic immunoassay for bovine growth hormone was

demonstrated. From aqueous chloro-aureic solutions, gold nanoparticles (size ~ 120 nm) were *in-situ* synthesised inside microchannels onto the PDMS surface via reduction of gold ions by the curing agent of the PDMS prepolymers. The gold nanoparticles were then activated with NHS-chemistry, and capture antibodies were attached. The assay read-out was done by localized surface plasmon resonance [359]. A microphysiological strain sensor for use in well plates to detect contractions of cardiac muscle tissues in culture to record drug induced stress was demonstrated, thereby expanding the application field of PDMS stretchable strain sensors to cell cultures [360].

3.5.9. Liquid alloys

Certain metals and some metal alloys are liquid at room temperature and do not exhibit the toxicity and environmental concern of mercury. They are suited for the creation of elastic conducting objects in combination with suited carrier materials, as recently reviewed [361]. Filling of dead-end microfluidic channels in PDMS of up to 1 m length and down to several μm in width with Gallium/Indium alloy was presented *via* application of a vacuum, which takes advantage of the permeability of PDMS to air [362]. Gallium-Indium-Tin (GalInSn) alloys like Galinstan[®] (exact composition is not publicised) can be liquid at room temperature and are suited to replace the toxic mercury historically used for such liquid metallic contacts. A PDMS surface was covered by a copper foil acting as stencil and a screen of Galinstan[®] was dispensed on top. Removal of the stencil and coverage with a second layer of PDMS produced a flexible microfluidic electronic device, as shown with the application as RFID (radio frequency identity) [363]. See Fig. 13.

Galinstan[®] has the disadvantage of oxidizing. To reverse this process, a PDMS device for the reduction of oxidized Galinstan[®] was demonstrated. Parallel microchannels filled with Galinstan[®] and HCl with a distance of 0.2 mm were employed, taking advantage of the permeability of PDMS for HCl. HCl reduces the oxidized Galinstan[®], which was observed by means of the contact angle of Galinstan[®] with PDMS, and the shape of the droplet, which becomes spherical upon chemical reduction of its oxidized surface [364]. An all-soft battery-free, wireless sensor for volatile organic compounds in liquid and gas phase, based on liquid gallium indium alloy in PDMS microchannels was demonstrated and its performance compared to solid state PDMS film gas sensors. The readout is done *via* a soft readout coil, the measuring principle is an all-soft interdigitated inductive-capacitive (LC). This novel sensor is advertised to function as wearable device as part of the 'internet of things' (IoT) [365]. In another recent approach, strain sensor, interconnects, and antenna are made of liquid metal (GalInSn) embedded in PDMS, enabling a wireless (by near-field communication, which powers the sensor and reads it out), soft sensor with a diameter of 2 cm for monitoring of human motion, as demonstrated by real-time readout of multiple fingers [350]. Tubular microfluidics made of PDMS were developed by a custom-built continuous extrusion and curing process, allowing the production of microtubes of up to 50 cm length and with an inner diameter of down to 10 μm . This concept was used for different applications. i) When they are filled with eutectic gallium indium, they turn into a soft, stretchable, thin wire. The electrical resistance thereof changes upon mechanical compression, converting them into a touch sensor. Such tubular microfluidics were woven into a fabric, resulting in a fabric touch sensor which is imperceptible, and which was read out wirelessly, ii) they can be turned into a self-propelled microscale motor when filled with catalase and immersed into a solution of H_2O_2 , which leads to the generation and expulsion of oxygen iii) triboelectric currents were detected when flowing KCl into the microtubes [366]. Another work presented a flexible pulse sensor embedded in PDMS, capable of measuring the pulse accurately at various locations of the human body and sending it to a computer via attached

modules (Bluetooth module and Arduino board). The sensor was created *via* a selective liquid-metal plating process (SLMP), which includes a Cr/Au deposition and etching step, followed by liquid metal being plated into a circuit, which was equipped with electronic elements (LED, optical receiver, chips, diodes, resistors and capacitors) [367]. Microfluidic capacitive touch sensors and flexible interconnectors were achieved via freezing and soft lithography of liquid metal (eutectic Galn) [368]. An implantable liquid alloy filled PDMS neural microelectrode (0.5 mm thick) [369] was used to stimulate peripheral nerves in frogs.

3.5.10. Polymers

Polypyrrole electrodes on PDMS were used to build a flexible polymeric microelectrode array (MEA), with features such as high stretchability (100%), low Young's modulus (450 kPa), high recycling stability and good adhesion to substrate (PDMS). Polypyrrole has a good electrical conductivity, biocompatibility and has been extensively used in tissue engineering. The challenge is to achieve high stretchability of the electrode together with its strong adhesion to the substrate. Polypyrrole nanowires were fabricated, which served as a strong connector between PDMS substrate and electrode, which was then brought into contact with a mechanically prestretched PDMS substrate that was covered with a thin layer of non-cured PDMS serving as a glue. The concept was applied as an electrocorticograph (brought into contact with the brain) recording electrophysiological pulses of neurons, where it enabled the discrimination between signals originating from epileptic and healthy rats [370].

3.5.11. Composites

Composites can have the advantage of combining properties of different materials, thus rendering better suited materials, e.g. by reduction of their percolation limit. Conductive composite films of micrometre-sized silver flakes and MWCNTs were created, where the nanotubes act as connectors between the silver flakes. The maximum conductivities of the hybrid silver-nanotube composites were 5710 S cm^{-1} at 0% strain and 20 S cm^{-1} at 140% strain, at which point the film ruptured [371]. Others produced a transparent conductive film, here silver nanowires (30 nm diameter, 25 μm long) were spin-coated on a silane-modified PDMS surface, followed by spin-coating of graphene-oxide flakes (flake size $\approx 0.05\text{ mm}$). In this way, the low reactivity of PDMS surfaces could be overcome [372]. A transparent conductive electrode (TCE) was created based on a silver nanowire percolation network modified with graphene oxide. The graphene oxide sheets solder the silver nanowire junctions and thereby reduce the inter-nanowire contact resistance without heat treatment or high force pressing. The composite network has a sheet resistance of $14\text{ }\Omega/\text{sq}$ with 88% transmittance at 550 nm A stretchable white polymer LED was fabricated for the first time employing TCE as cathode and anode [373]. See Fig. 14. A simple and promising approach consists of a 3D-printed porous structure of PLA that was backfilled with PDMS, with subsequent dissolution the PLA template with dichloromethane. The PDMS was filled with a graphene oxide/ MWCNT solution. This produced a conductor also under strain (40% residual conductivity at 100% stretching) [139].

A flexible (up to 25% stretch) magnetic sensor based on the giant magnetoresistance effect (used for example in hard disk drives) was established. The sensor was fabricated via strain release patterning of a spin valve. A mask was used to produce parallel ribbons to reduce cracking in the lateral direction upon strain release. The concept is amenable to stretchable, wearable electronics [374].

A sensor for bending of fingers of a latex glove was demonstrated. Thermoset polymers such as a polyimide tape were selectively pyrolyzed by laser and then the carbonized trace was embedded in PDMS which was then cured and thus creates a conductive layer. Embedded carbonized materials were com-

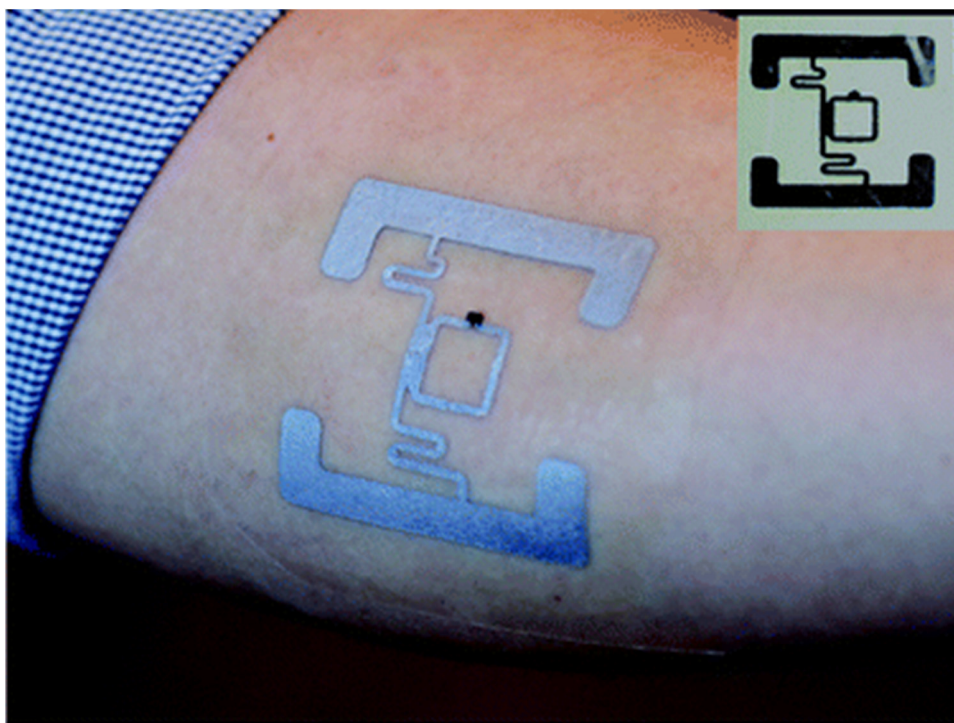


Fig. 13. Photo of the demonstrated stretchable/conformable UHF RFID tag [363].

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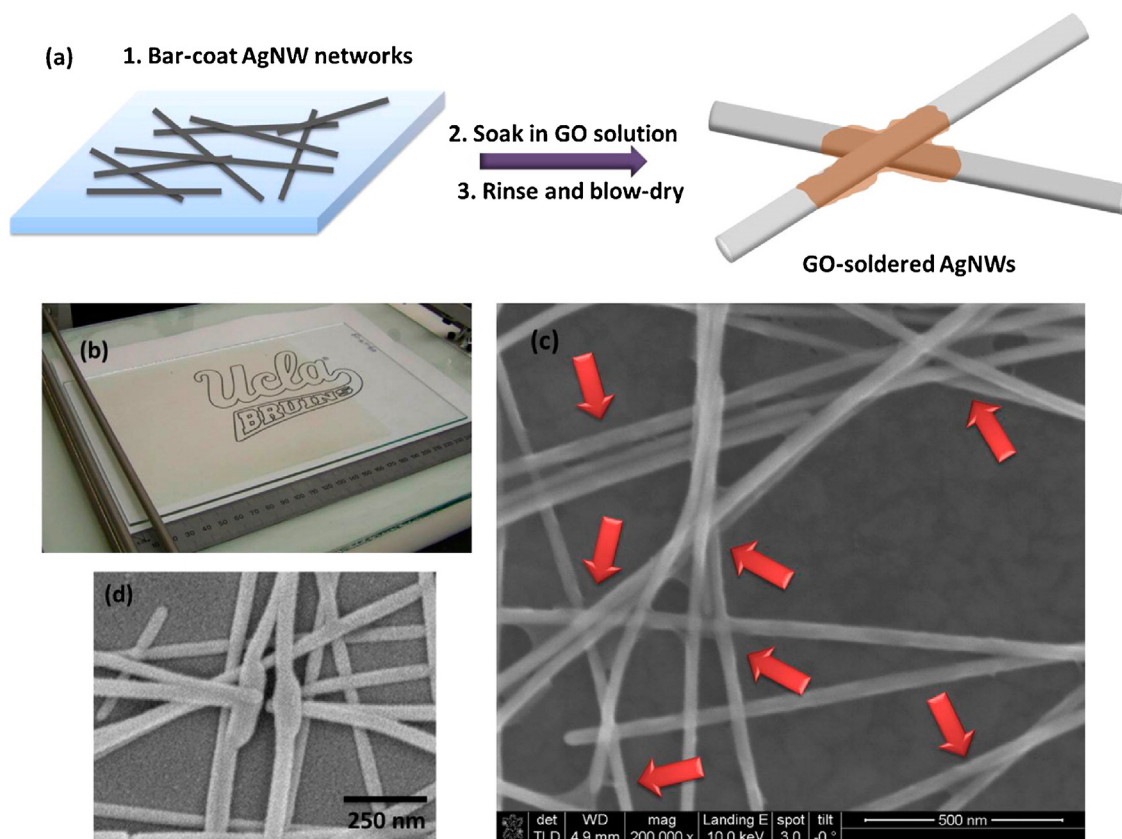


Fig. 14. (a) Schematic illustration of the fabrication of a GO-AgNW network on a glass substrate at room temperature. (b) Optical photograph of a AgNW network (20 × 15 cm²) bar-coated on a drawdown machine. SEM images of (c) GO-soldered AgNW junctions (indicated by red arrows) and (d) typical high-temperature fused AgNW junctions [373]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

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posed of partially aligned graphene and carbon nanotube particles, with a sharp directional anisotropy, enabling unidirectional strain sensors. The sheet resistance is as low as 60 Ohm/square and extraordinary gauge factors of up to 20,000 by 100% stretching were achieved [315].

Thus, novel characteristics were demonstrated, overcoming limitations in resistance and gauge factor as published so far. A spider silk-like (SSL-SWCNT) layer with superior strain durability and resistivity was deposited onto a prestrained PDMS surface *via* transfer from an aluminium foil, creating buckles when released. Subsequently, a gold layer (200 nm) was sputtered onto this surface. Upon stretching of the PDMS surface, cracks, similar to the spider slit organs, appear. The SSL-SWCNT was demonstrated to measure the cardiac pulse wave when attached to the wrist. It can also function as a temperature sensor [375]. CNT-silver nanoparticle (CNT-AgNP) composites were painted (resulting in 2 μ m thickness) on an array of parallel PDMS fibres (manufactured *via* moulding in Si), with the top side as a strain sensor [5 wt% AgNP] and the bottom side as an interconnect electrode [30 wt% AgNP], to form electronic whiskers (*ë*-whiskers; width 250 μ m, length 15 mm). The AgNPs serve to decrease the CNT-CNT junctions' resistivity. Gauge factors of up to 95 were achieved to measure wind flow with a sensitivity of 8%/Pa [376]. A similar approach was published shortly thereafter, with a higher sensitivity of 59%/Pa [377].

3.6. Magnetic

Magnetic layers are an attractive solution to enable electromechanical actuation in microsystems.

A thin magnetic film of alternating layers of NdFeB and Ta was sputtered onto a silicon substrate, as the PDMS would not withstand the high sputtering temperature. PDMS was cast on top of the silicon substrate. After curing, the silicon was etched away with XeF₂ gas, thereby leaving the PDMS with the magnetic film. The design is proposed to be used for the creation of microrobots [378].

Conversely, systems combining PDMS with magnetic layers have been shown to be candidates for the mechano-electric transduction, e.g. in the rapidly growing field of environmental energy harvesting for miniaturized electronics. To this purpose, a triboelectric nanogenerator (TENG) approach was demonstrated with a layered setup, a magnetic response layer (PDMS / Fe-Co-Ni powder composite) and an arced indium tin oxide (ITO) / polyethylene terephthalate (PET) film. The ITO/PET film is used as induced charge collector, electrode, and supporting layer. By mechanical motion of an external magnet, the PDMS layer will contact and separate with the ITO film repeatedly, thus generating an alternating-current (AC) power [379].

3.7. Structural modifications

3.7.1. Surface structure modifications

3.7.1.1. Physical properties. Mechanical stretch of PDMS with grooves generated tuneable wetting (isotropy vs. anisotropy) just by applying stretch in two orthogonal directions (parallel and perpendicular to the grooves) [380]. In a similar design, the deposition of Si nanoparticles onto the elevations of PDMS grooves, the anisotropic wetting (non-circular wetting profile in *x* and *y* of sessile drops) could be altered unidirectionally along the grooves [381]. Anisotropic wetting was also achieved by deposition of silver nanorods onto a pre-stretched (30%) PDMS surface resulting in an anisotropy of contact angle with water of 37° [382]. Another example is the roughening of PDMS surfaces, which was achieved by applying a simple, fast, and inexpensive method using a aqueous slurry of alumina particles. The surface roughness was measured by AFM and the chemical alterations were detected by XPS. The application of this method for microcontact printing of

BSA-Alexa647 showed improved transfer from roughened PDMS surfaces to roughened photoresist surfaces SU8 and 1002 F [383]. A super-hydrophobic surface was created by micro/nano dual-scale hierarchical structures (WCA >160°), that can be applied to render a surface self-cleaning. The surface structures were created by a single-step process using a reusable silicon mould that is covered with a fluorocarbon layer [384]. Polychlorinated biphenyls (PCBs) were detected via capture with an aptamer with high sensitivity on a PDMS surface by surface enhanced Raman spectroscopy. The PDMS surface was decorated with an Ag nanocrown pattern, which was replicated from an anodic aluminium oxide template [385]. With the goal of producing a tactile skin, a pressure sensor with pyramid shaped features was produced by moulding of a silicon master. On top of the surface, an indium tin oxide coated conductive flexible sheet of poly(ethyleneterephthalate) (PET) was laminated [386]. Rough PDMS surfaces comprising 3 μ m hemispherical bumps and cavities with pitches ranging from 4.5 to 96 μ m and were fabricated by moulding and a smooth served as control. Their wetting and dewetting behaviour with water was investigated for the understanding of print surfaces used for example in printed electronics. It was found that the static contact angles for the sessile water droplets on smooth, bumpy, and cavity PDMS surfaces were 110°, 122°, and 124° (for 12 μ m pitch length), respectively. They were all found to be in their metastable wetting states because these static contact angles are all converted to ~91° after vibrating the sessile droplets for 15 s [387].

3.7.1.2. Cell culture. Micro- and nano-patterning of PDMS can be employed to study cell/structure interaction, as was reviewed [388]. Nanopatterned gratings with 350 nm width and 350 nm depth were used as a substrate for smooth muscle cells. The cells cultured on the gratings showed an elongation similar to the physiological shape, however, the proliferation was reduced [389]. Human mesenchymal stem cells behaved differently on rigid PDMS with 350 nm gratings but not on soft planar PDMS [390]. Macrophage adhesion is desirable on implants, and its reduction on PDMS with gratings of 500 nm was most pronounced [391]. A gradient of varying amplitude and wavelength of grooves in PDMS was introduced via air plasma treatment under strain with a shielded oxidation triangular prism mask. Culturing of osteoblast cells showed alignment with most pronounced alignment at amplitude 130–180 nm and wavelength 550–730 nm [392]. PDMS that was micropatterned (lines 750 nm spacing and width, pillars) with varying elasticity (500–2000 kPa) was oxidized with oxygen plasma, coated with fibronectin, and incubated with 3T3 fibroblasts and cancer fibroblastic cells (Sal/N). 3T3 polarisation and spreading are influenced by microtopography (lines) and elasticity, the cancer cells react only to the topography (lines) but not to elasticity [393].

It is sometimes of interest to modify a PDMS surface, both with or without changing its chemical properties. The easiest way is to stretch the PDMS sample, deposit a layer of metal, or simply treat it with air plasma, and release it, thus inducing wrinkles with a defined length scale and amplitude to the surface [394]. Another straightforward approach was to heat the PDMS, leading to thermal expansion, then to deposit Ti and Au from vapour phase. Upon cooling a buckled surface appeared [352]. The elasticity of PDMS was varied by a temperature gradient during curing. The deposition of nickel produced varying surface structures relating to the elasticity of the PDMS substrate, such as folds, nested pattern, wrinkles [395]. *N*-isopropylacrylamide (NIPAM) concentration gradients in water were utilized in a microfluidic setup to create surface roughness gradients with a range of up to ~163 nm. NIPAM was “grafted onto” the PDMS surface via UV radiation. The number of attached cells were reduced with increase of surface roughness [396]. A 3D PDMS scaffold microfluidic chip was created from a Ni foam substrate for the isolation of circulating tumour cells from the blood

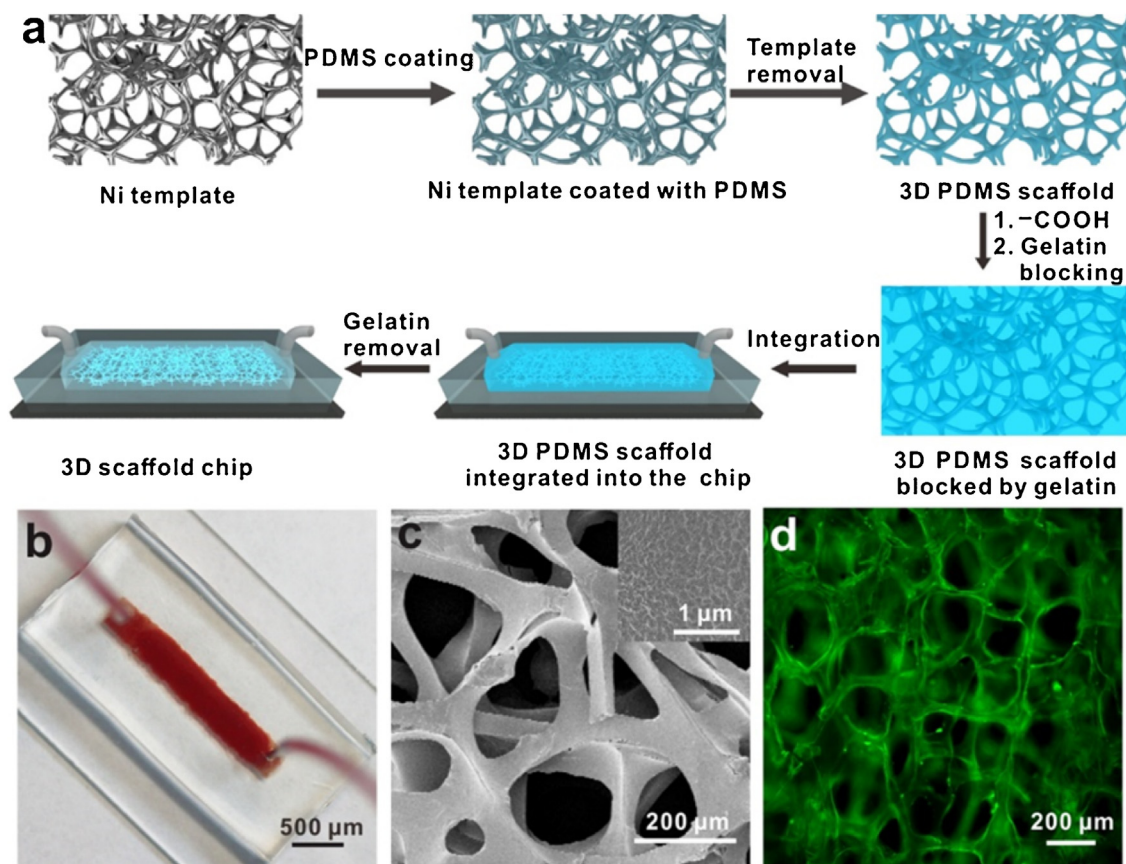


Fig. 15. (a) Procedure for preparation of 3D scaffold chip, (b) blood stuffed 3D scaffold chip, (c) SEM image of 3D scaffold (inset, magnified SEM image of PDMS scaffold), (d) characterization of anti-EpCAM antibody-functionalized 3D scaffold chip by FITC-labeled goat antimouse IgG [397].

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via antibody capture. The antibodies were immobilized via plasma treatment followed by immersion in carboxyethylsilanetriol and NHS chemistry [397]. See Fig. 15.

3.7.2. PDMS foams (PDMS sponges)

Foam structure of polymers leads to major changes in mechanical properties surface area accessible to adsorption. Recently, a review on PDMS foams (also termed PDMS sponges) was published [398]. Several approaches have been proposed to produce PDMS with sponge structure.

Removal of oil spills in sea water by sorbents is an ongoing topic of research [399]. Widely used sorbent materials include the following: zeolites, carbon nanotube (CNT) sponges, Fe/CNT composites, carbon fiber aerogel, steelmesh, and PDMS coated polyurethane (PU) sponge [400]. Ideally, such a sponge or foam should be hydrophobic (not absorbing water) and oleophilic (readily absorbing oil).

Various PDMS foams (also termed PDMS sponges) have been developed in order to remove organic compounds, such as petroleum products from water. An Au nanoparticle-PDMS nanocomposite foam was established by stirring using water as medium. The particles had a dimension of 10–50 nm [401]. The foam shows high swelling for organic solvents such as xylene, benzene and other organic solvents with a swelling ratio of up to 600%. Heating the foam at 300 °C in air for 30 min, restores its original state. Initially, the heating of plain PDMS foam shows a gradual weight loss up to 20%, while heating, which is attributable to the depolymerization of PDMS chains to give volatile cyclic oligomers [402]. This approach does not seem to be readily amenable when trying to absorb large quantities of oil from sea water, due to the

necessity of heating the composite to remove absorbent before its reuse.

The sugar template method is an easy way to produce a PDMS sponge. A comestible sugar cube (composed of saccharose) was placed in PDMS prepolymer mixture, upon which absorption via capillary forces occurred. After curing at 120° for 12 min, the sugar was removed from the composite by dissolution in a sonicated water bath at 40° (duration not mentioned). The foam readily absorbed oil layers and hydrophobic toxins on water when floating in them. For example, the retention rate of 1,2-dichlorobenzene (a toxic organic contaminant in water) was 1000 wt%. The foam can be mechanically squeezed and reused [403]. See Fig. 16.

An alternative sugar template method involves PDMS prepolymer that were mixed with p-Xylene and crystalline sugar particles of varying diameter. The composite was cured at 70° for 12 h. The sugar was removed by soaking in water at 80° (duration not mentioned), and then in ethanol to remove the p-Xylene. The WCA was increased up to 144°. The foam could absorb 20 × w/w petroleum ether. This approach has higher absorption values than the sugar template method mentioned above, but involves more steps and organic solvents. The absorbed oil can be collected by squeezing [404].

A porous PDMS-graphene composite sponge which had both hydrophobic and oleophilic properties was created using the above-mentioned sugar template method. Graphene sheets were dispersed in DMSO, which was injected to the dry sponge and let to dry. This was repeated three times. The graphene sheets increased the porosity and the WCA of the sponge (from 102° to 127°). When an oil drop was dispensed on top of the graphene sponge,

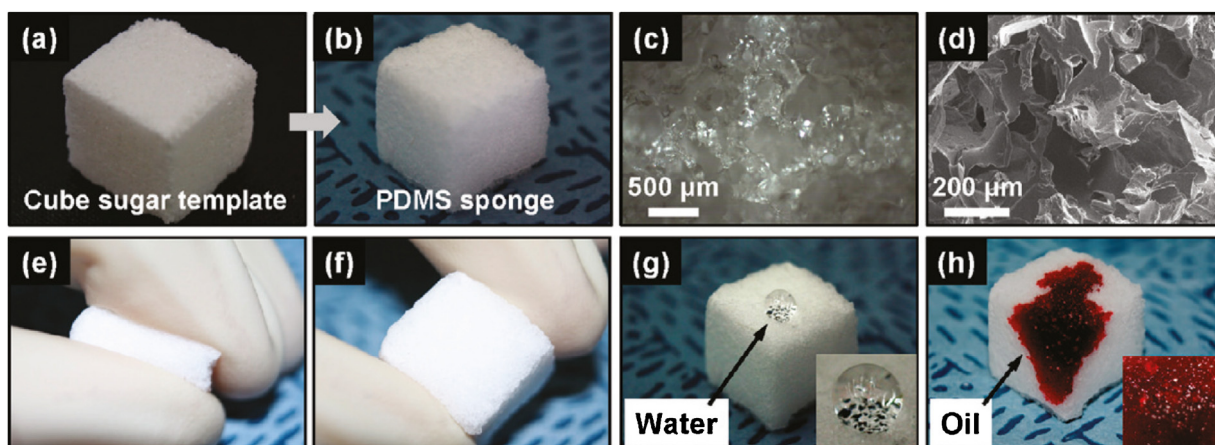


Fig. 16. (a) Photograph of a commercially available cube sugar for the templating process. (b) Photograph of the as-replicated PDMS sponge. (c) Optical microscope image of the cross-sectional area of the PDMS sponge. (d) SEM image of the cross-sectional area of the PDMS sponge. (e) Manual compression to over 50% reduction in volume and (f) recovery to the original shape. PDMS sponge wettability showing (g) high hydrophobicity and (h) strong oleophilicity of the surface. Transformer oil was used and colored with Oil red O dye for clarity. (Inset) Magnified view of water and oil droplets on the PDMS sponge [403]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

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it was adsorbed instantly. For toluene, up to 800 wt% absorption was achieved. Reuse was also possible by squeezing [405].

An alternative approach to manufacture a PDMS sponge was reported by using citric acid monohydrate as a hard template being mixed with prepolymers, which subsequently were cured. Extraction in ethanol as a solvent for 6 h followed. The authors claim shorter extraction duration than with sugar templates. The WCA of the sponge was 139° [406]. Another published work also used citric acid monohydrate as template, in this case to form a CNT/PDMS sponge nanocomposite. The composite was applied as a strain sensor (gauge factor 15, strain >15%) [319].

A lightweight, flexible PDMS foam/graphene composite was produced as a electromagnetic interference shielding (EMI). The PDMS base, curing agent, and ethyl acetate were mixed at 10:1:100, cast on graphene (grown on a nickel foam), and cured at 100° for 1 h. The nickel foam substrate was etched away, with HCl, leaving a highly porous PDMS-graphene ensemble, with an electrical conductance of 2 S cm⁻¹ with a graphene loading of ~0.8 wt%. In summary, PDMS foams are easy to manufacture and expose pertinent characteristics of PDMS like elasticity, strong and self-recovering hydrophobicity, low surface energy, chemical and environmental inertness, on a very large surface.

A method to create PDMS foams with varying foam density (0.08–0.46 g cm⁻³) and open or closed foam cells was presented. It relies on a single newly developed Piers-Rubinsztajn reaction [407].

A method that precludes the use of hard templates was presented recently. Basically, PDMS prepolymers were mixed with hexane and water to create an emulsion, with water acting as a soft template. The emulsion was cured at 120 °C, leading to the evaporation of water and the creation of large interconnected porous cavities (406 ± 302 μm). The advantage of this method is the rapid generation of large quantities of reusable PDMS foams to remove oil spills from water [408].

This yields yet another interesting, although not yet widely exploited, building block for future microsystems.

4. Permeability

PDMS is permeable to oxygen, CO₂ (and other gases) [409]. The diffusion coefficients for oxygen and CO₂ in PDMS are ~2000–4000 μm² s⁻¹ [49] and ~1000 μm² s⁻¹ [50] respectively (compared to 2100 μm² s⁻¹ and 1920 μm² s⁻¹ in water [410]). Dif-

fusion of oxygen was reduced after plasma treatment of PDMS due to the formation of a silica like surface. This effect disappeared after three days when stored in air (3 weeks when stored in water) [411]. The permeability of PDMS to air could be tuned by using different mixing ratios of the prepolymers [412]. For application in permselectivity (air separation, petrochemical separation), the permeability of PDMS membranes (200 nm to 1 mm) to He and CO₂ was measured. For thicknesses <200 nm, scanning electron imaging revealed pinhole defects, which led to changes in permeability and selectivity (He vs. CO₂) [413]. The creation and applications of membranes made of PDMS, allowing control e.g. of diffusion, was reviewed recently [414].

PDMS is also permeable to organic solvents and hydrophobic molecules (see also paragraph above: shortcomings of PDMS). PDMS swelling occurs with diisopropylamine, triethylamine, pentane, and xylenes, but not with water, nitromethane, dimethyl sulfoxide, ethylene glycol, perfluorotributylamine, perfluorodecalin, acetonitrile, and propylene carbonate. Solvents that swell the PDMS can be used to extract contaminants from the PDMS bulk [30]. Permeability of hydrophobic molecules can be a problem for example, when the assay analyte inside microchannels made of PDMS is hydrophobic. This could lead to changes in the measured concentration of compounds and aberrant behaviour of cells especially in micro-scale culture when the compound has an effect on cells (see also section in the introduction) [31–34]. The permeability of various hydrophobic molecules, such as rhodamine G, dexamethasone, diazepam, and other was quantified and their dependency of hydrophilic/hydrophobic balance was shown [31]. For example, estrogen concentration inside microchannels reduced to 10% after 24 h [32]. This potential confounder asks for dedicated surface modification of PDMS to retain correct assayed concentrations

The diffusion of a small organic molecule (rhodamine B) was reduced when the PDMS surface was treated with concentrated sulfuric acid. This treatment produced a thin brittle layer, that was characterized by SEM (scanning electron microscopy), EDX (energy-dispersive X-ray spectroscopy), ATR-IR (attenuated total reflection FTIR spectroscopy), and XPS, indicating a process involving a segmentation of the polymer backbone while sulfuric acid stabilizes the terminal silanol groups by forming organosulfates (by esterification of terminal silanol groups) [415]. Carbon nanomembranes (CNM) can be manufactured by crosslinking a monolayer of molecules containing phenyl moieties via electron exposure. The

resulting CNM has a thickness of 1 nm, and 3 nm for films with three layers. The permeability for various gases was measured, and plotted according to their kinetic diameter. Such thin carbon-based nanomembranes can be used as separation media for chemical and gas purification [416].

Initiated chemical vapour deposition (iCVD) was used to form a poly(1H,1H,2H,2H-perfluorodecylacrylate) (pPFDA) polymer layer on a PDMS mould (containing the microchannels) and on a PDMS covered glass lid. Like the initiators used in traditional free-radical polymerization, the initiators in iCVD are thermally-labile species that fragment to form radicals at moderate temperatures. Mould and lid were then brought into contact and bonded together using UV-photopolymerization of a mixture of 95 wt % polyethylene glycol diacrylate (PEG-DA, $M_w = 575$) and 5 wt % 2,2-diethoxyacetophenone (DEAP). The pPFDA polymer layer effectively reduced ambient oxygen into and organic solvent diffusion out of the microchannels. This construct was applied to synthesise microparticles of 3 μm diameter by radical polymerisation, where quenching by oxygen radicals is a well-known and limiting factor. This synthesis is reported to be extremely challenging inside non-modified PDMS microchannels, where the diffusion of oxygen is intense [417].

A simple method to tune PDMS air permeability is the variation of the mixing ratio of oligomer and curing agent, with 20:1 showing the highest (3 times higher than 10:1) and 5:1 showing the lowest permeability [412].

Diffusion of toluene into bulk PDMS from a microchannel was abrogated by a caulking silicate surface coating. Basically, the hydrophilic (WCA reduced from 101° to 77°) 400 nm thick silicate layer was created from polyvinylsilazane via UV- and thermal curing and subsequent hydrolysis in alkali solution [29].

As mentioned above, diffusion of water vapour through PDMS membranes can be a problem in cell culture in miniaturized dimensions with (sub-)microliter volumes, as evaporation of water increases the osmolality of the cell culture medium, thereby inhibiting cell growth and changing cellular behaviour. It was demonstrated that a PDMS-parylene-PDMS film could suppress evaporation [41]. Octadecyltrichlorosilane-modified PDMS was used to inhibit oligomer diffusion into the bulk PDMS in a nano-electrospray emitter [418].

A PEG (poly-oligoethylene glycol methyl ether methacrylate [POEGMA]) brush was grafted from a PDMS covered porous polyacrylonitrile membrane via ATRP. The WCA decreased from 108° to 70° .

To simulate the membrane surface, a silicon wafer was covered with 10% POEGMA and subjugated to CO_2 , which changed the conformation of the PEG chains, i.e. they were swelling like in a solvent. This concept may be applied as gas separation system [419].

A PDMS layer (65 or 200 μm) sandwiched between two support layers of high-porosity polyethylene and perforated metal was constructed for the pervaporation (permeation and evaporation) of 1-butanol, a biofuel alternative to gasoline or as a diesel additive [420]. In a similar manner (incorporating a 160 μm thick PDMS membrane) perstraction (permeation and extraction) was applied to separate butanol, acetone and ethanol from aqueous solutions using different hydrophobic ionic liquids [421].

PDMS has been used extensively in medicine, for example as packaging material for long-term, implantable electronic devices (e.g. pacemakers), blood pumps, mammary prostheses, catheters, shunts, cochlear implants, and oesophagus replacements [422].

In implantable electronic devices, permeability control is critical. For many implants (e.g. cardiac pacemakers), Ti is used as an encapsulant. If RF communication with external devices is required, glass may be applied as encapsulant.

Water vapour transmission rates in polymers is several order of magnitudes higher than in Ti or glass encapsulants. The high gas

permeability of PDMS therefore prevents it from being used as hermetic packaging, as it would lead to high humidity in the implant, which can disable or damage the electronic circuitry. To solve this issue, a siloxane/parylene-C multilayer packaging reduced the water vapour transmission rates [423].

5. Smart, functional, reactive materials based on PDMS

Recently, significant progress has been achieved in smart, functional and reactive materials based on various base chemistries functional groups, targeting many application fields. A smart material can be defined as a material that reversibly alters its properties based on a given stimulus. A comprehensive review of this burgeoning field at this place is out of scope, but we mention a few representative examples here. Polymer mechanochemistry, embedding of so-called mechanophores into polymers such as PDMS, can be used to produce a strain-induced colour change via covalent bond activation. Covalent incorporation of the strain responsive molecule bis-alkene functionalized spiropyran into PDMS (0.5–0.7 wt%) was done via platinum cure hydrosilylation, the same reaction used to form the PDMS network. Spiropyran reversibly converts to colored merocyanine under strain and thereby allows for a functional material with direct spatial and temporal visualization of stress accumulation [424]. Piezoelectric harvesting of mechanical stimuli can be applied to wearable electronics, where mechanical energy is transformed into electrical power. An example is the use of native cellulose microfiber and PDMS as a substrate and MWCNTs as conductive filler [425]. See also the chapter 2.4. on energy harvesting. An octopus-inspired smart adhesive pad was demonstrated. A thermoresponsive hydrogel layer acts as artificial octopus sucker, which is embedded in PDMS. This approach enabled smart printing of an InGaAs array onto an SiO_2/Si substrate. However, the smart-printing process is applicable to various micro-/nanostructured arrays on arbitrary rigid or flexible substrates [426]. Superhydrophobic films (usually achieved by combination of low surface energy and presence of surface roughness [183]) made of $\text{TiO}_2\text{-SiO}_2$ @PDMS were created (WCA > 160°). Silica and titania gels were functionalised with methyl groups by adding hexamethyldisilazane (HMDS) and then blended with a PDMS matrix. The films were created by sol-gel processing and became superhydrophilic (WCA < 5°) through calcination when heated to 470°C via replacement of Si-CH_3 by Si-OH . Coating of polyester-cotton fabrics rendered them into a superhydrophobic mesh for oil-water separation. Photocatalytic activity of $\text{TiO}_2\text{-SiO}_2$ @PDMS was successfully applied to the breakdown of methylene blue [427]. In a similar approach a composite of PDMS and SiO_2 nanoparticles (16–20 nm diameter) was spray coated onto glass slides, which resulted in a superhydrophobic WCA of 153° . Thermal calcination at 500°C oxidized the methyl groups and rendered the surface superhydrophilic (WCA 0°). Nanometre and micrometre surface roughness of PDMS/SiO_2 is responsible for the surface superhydrophobicity [428]. These diverse examples demonstrate the potential of such responsive polymers in a variety of applications. Numerous other examples highlighting the potential of PDMS as basis for 'smart' materials are interspersed in other chapters in this review.

6. Summary, open questions and directions for future research

This review summarizes the recent research efforts done in surface and bulk modifications of PDMS with an emphasis on their utility for microfluidic systems. Some techniques that are currently not directly related to microfluidics were included, as they may have potential impact on microfluidics in the future. Also, 'hot'

topics of functional PDMS were included, especially regarding flexible electronics. PDMS has advantages such as biocompatibility, transparency and gas permeability. However, PDMS also has limitations (with regard to the application), such as hydrophobicity (and hydrophobic recovery after oxidation of its surface), fouling (the adsorption of compounds, bacteria and cells), and permeability to small hydrophobic molecules (which limits its use in their detection and the validity of cell culture experiments), and its undisclosed composition (Sylgard 184®). Much effort has been paid to the generation of hydrophilic surface modifications, with mixed success regarding stability. More research needs to be done regarding this issue. A hot topic is the achievement of electrically conducting PDMS, responding to a need for Adv Mater for flexible, transparent electrodes for flexible electronics and microfluidics. Electrodes may have stable or variable resistance upon stretching, in the latter case allowing their use as strain sensors. Surface modification of PDMS with metals is not straightforward, requiring special techniques. Recently, graphene and carbon nanotubes have raised special interest, due to their unique characteristics, such as a high conductivity and mechanical hardness, leading to composite PDMS materials with outstanding characteristics: flexible electrodes can be created by spray-coating of SWCNTs onto PDMS, and composites of carbon nanotubes with silver flakes were demonstrated to work as strain sensors. Also, flexible electronic microfluidics made of liquid alloys were presented recently.

While the 'static', constitutive properties of PDMS have been a driver for its use in the past decades, there is currently a shift in research towards designing active, functional, reactive PDMS systems with advanced functionality. Such sensors, transducers, actuators and combined systems that often exploit properties of PDMS arising by hybridization with other materials render the field of PDMS modification a hot and dynamic topic in current research and lead to very valuable, novel engineering materials.

Adv Mater based on modifications of PDMS, in bulk or on its surface, are therefore candidates to consider wherever challenging problems in microsystem design call for innovative solutions.

Regarding commercialisation of products made of PDMS there are several limitations, such as its undisclosed composition, its high price (~150\$/kg as compared to ~3\$/kg for polystyrene [PS] or poly(methyl methacrylate) [PMMA]), and the low scalability of soft lithography (due to time-consuming casting process of 0.5–1 h for one cycle) as compared to injection-moulded, hot embossed, or roller imprinted thermoplastics, such as PS, or PMMA [429]. This makes PDMS ideal for low-volume production for research labs to be used as prototypes for proof-of-concept studies. For certain biological applications the absorption of external and leaking of internal molecules and the solvent swelling behaviour can be a reason to opt for other materials, although coating strategies to solve these problems exist. Other silicone products (besides Sylgard) from DOWSIL have been approved for medical products, such as electronic implants, catheters, contact lenses, breast implants, among other, proving its suitability in this field. In summary, currently there are new, burgeoning fields of research using PDMS, such as anti-fouling, self-healing, energy harvesting, dry adhesives, (wearable) flexible electronics, e-skin, oil-spill decontaminants, etc., keeping it a versatile tool for cutting edge research and applications design.

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